actuarial-science: the Actuarial Science and Insurance program, user guide

actuarial-science-ai999x 1.6.0, Rust build: command line, desktop app and browser. **Creator / engineer:** Ibnul Jaif Farabi · **Company:** AI999X · support@ai999x.com

Actuarial Science and Insurance covers life tables and survival models, life insurances and annuities, net and gross premiums, reserves and Thiele's equation, interest theory, loss models with deductibles and limits, Panjer's recursion, VaR and TVaR, credibility and loss reserving (chain-ladder, Bornhuetter-Ferguson, Cape Cod, Mack). This guide is about `actuarial-science`, the model as a self-contained program for Windows and Linux: 93 capabilities in 15 topics, answered on your own computer with no network, no Python and nothing else to install. The same compiled model runs in the AI999X desktop app and in the browser on the model's page (https://www.ai999x.com/models/actuarial-science-ai999x/); the Python package (`pip install actuarial-science-ai999x`) has its own guide. Every command output printed here is what the program printed.

Contents

- 1. Get it: desktop app, command line, browser, Python
- 2. Quick start
- 3. Writing inputs
- 4. Outcomes and exit codes
- 5. Scripts and agents: batch and JSON
- 6. Figures
- 7. In a web page or Node.js: WebAssembly
- 8. Every capability

1. Get it

Version 1.6.0 of the AI999X desktop app and of the command-line bundle. Checksums for every file: https://downloads.ai999x.com/desktop/1.6.0/SHA256SUMS; the newest version is always under https://downloads.ai999x.com/desktop/latest/. Full instructions: https://www.ai999x.com/desktop-cli/

what	system	download
Desktop app: Actuarial Science and Insurance	Windows x86-64	https://downloads.ai999x.com/desktop/1.6.0/ai999x-actuarial-science-1.6.0-windows-x86_64.zip
Desktop app: Actuarial Science and Insurance	Linux x86-64	https://downloads.ai999x.com/desktop/1.6.0/ai999x-actuarial-science-1.6.0-linux-x86_64.tar.gz
Desktop app: all models	Windows x86-64	https://downloads.ai999x.com/desktop/1.6.0/ai999x-1.6.0-windows-x86_64.zip
Desktop app: all models	Linux x86-64	https://downloads.ai999x.com/desktop/1.6.0/ai999x-1.6.0-linux-x86_64.tar.gz
Command line	Windows x86-64	https://downloads.ai999x.com/desktop/1.6.0/ai999x-cli-1.6.0-windows-x86_64.zip

what	system	download
Command line	Linux x86-64	https://downloads.ai999x.com/desktop/1.6.0/ai999x-cli-1.6.0-linux-x86_64.tar.gz

Desktop app

Actuarial Science and Insurance in its own app: search, questions in words, input forms, live figures and explanations. Windows: Download ai999x-actuarial-science-1.6.0-windows-x86_64.zip, right-click it → Extract All, then open the folder and start ai999x-actuarial-science.exe. If Windows SmartScreen asks, choose More info → Run anyway. Linux:

```
curl -LO https://downloads.ai999x.com/desktop/1.6.0/ai999x-actuarial-science-1.6.0-linux-x86_64.tar.gz
curl -LO https://downloads.ai999x.com/desktop/1.6.0/SHA256SUMS
sha256sum -c --ignore-missing SHA256SUMS
tar xzf ai999x-actuarial-science-1.6.0-linux-x86_64.tar.gz
./ai999x-actuarial-science-1.6.0-linux-x86_64/ai999x-actuarial-science
```

It opens on this model: type a question or pick a capability, fill in the form and press Compute. The AI999X desktop app with every model works the same way (choose the model first). Windows: Download ai999x-1.6.0-windows-x86_64.zip, right-click it → Extract All, then open the folder and start ai999x.exe. If Windows SmartScreen asks, choose More info → Run anyway. Linux:

```
curl -LO https://downloads.ai999x.com/desktop/1.6.0/ai999x-1.6.0-linux-x86_64.tar.gz
curl -LO https://downloads.ai999x.com/desktop/1.6.0/SHA256SUMS
sha256sum -c --ignore-missing SHA256SUMS
tar xzf ai999x-1.6.0-linux-x86_64.tar.gz
./ai999x-1.6.0-linux-x86_64/ai999x
```

Command line, Linux

The bundle holds one program per model; `actuarial-science` is this one. The checksum line should print OK.

```
curl -LO https://downloads.ai999x.com/desktop/1.6.0/ai999x-cli-1.6.0-linux-x86_64.tar.gz
curl -LO https://downloads.ai999x.com/desktop/1.6.0/SHA256SUMS
sha256sum -c --ignore-missing SHA256SUMS
mkdir -p ~/.local/share && tar xzf ai999x-cli-1.6.0-linux-x86_64.tar.gz -C ~/.local/share
echo 'export PATH="$HOME/.local/share/ai999x-cli-1.6.0-linux-x86_64:$PATH"' >> ~/.bashrc && source ~/.bashrc
actuarial-science --version
```

Command line, Windows (PowerShell)

The fourth line should print True.

```
$F = "ai999x-cli-1.6.0-windows-x86_64.zip"
Invoke-WebRequest "https://downloads.ai999x.com/desktop/1.6.0/$F" -OutFile $F
Invoke-WebRequest "https://downloads.ai999x.com/desktop/1.6.0/SHA256SUMS" -OutFile
SHA256SUMS
(Get-FileHash $F -Algorithm SHA256).Hash.ToLower() -eq ((Select-String -SimpleMatch $F
SHA256SUMS).Line -split ' ')[0]
Expand-Archive $F -DestinationPath "$env:LOCALAPPDATA\AI999X" -Force
$D = "$env:LOCALAPPDATA\AI999X\ai999x-cli-1.6.0-windows-x86_64"; $env:Path += ";$D"
[Environment]::SetEnvironmentVariable("Path", [Environment]::GetEnvironmentVariable("Path",
"User") + ";$D", "User")
actuarial-science --version
```

In your browser

Open https://www.ai999x.com/models/actuarial-science-ai999x/, choose a capability and press Compute in your browser: the same program runs on your device as WebAssembly; nothing is sent anywhere.

Python

```
pip install actuarial-science-ai999x
```

2. Quick start

2.1 Check the install

Prints the model, its version and that this is the Rust build.

```
$ actuarial-science --version
actuarial-science-ai999x 1.6.0 (Rust build, ax-engine 1.6.0)
```

2.2 Find a capability

Search the capabilities by what you want to know; the first column is the id to pass to info and solve.

```
$ actuarial-science search "net annual premium for whole life insurance"
act_net_premium_whole_life           Net annual premium for whole life insurance  (premiums
/ net premiums)
act_net_premium_limited_pay          Net annual premium for limited-payment whole life
insurance  (premiums / net premiums)
act_reserve_whole_life               Net premium reserve for whole life insurance
(reserves / prospective reserves)
act_gross_premium                    Gross annual premium for whole life insurance with
expenses  (premiums / gross premiums)
act_makeham_continuous_whole_life    Continuous whole life insurance, premium and reserve
under Makeham's law  (reserves / Thiele's differential equation)
act_net_premium_term                 Net annual premium for term insurance  (premiums / net
premiums)
act_net_premium_endowment            Net annual premium for endowment insurance  (premiums
/ net premiums)
act_gross_premium_reserve            Gross premium reserve for whole life insurance with
expenses  (reserves / gross premium reserves)
… 12 more
```

2.3 List the capabilities

Every capability the program answers: its id (what you pass to solve) and its name.

```
$ actuarial-science list
act_accumulated_value                   Accumulated and present value of a single amount
act_annuity_certain                     Annuity-certain present and accumulated values
act_annuity_yield                       Yield rate of a level annuity
act_bornhuetter_ferguson                Bornhuetter-Ferguson reserve
act_buhlmann_credibility                Buhlmann credibility premium (known structural
parameters)
act_buhlmann_empirical                  Empirical Buhlmann credibility from a data table
act_buhlmann_straub                     Buhlmann-Straub credibility with exposures
act_cape_cod                            Cape Cod (Stanard-Buhlmann) reserve
… 85 more
```

2.4 What a capability needs

The equation or method, every input and output with its SI unit. Read this before solve: it tells you the input names.

```
$ actuarial-science info act_net_premium_whole_life
Net annual premium for whole life insurance
  equivalence principle: P ä_x = S A_x
  in   q        death probabilities q_x, q_(x+1), ... ending with 1 (a list) (vector) []
  in   i        annual effective interest rate (5% or 0.05) []
  in   S        sum insured []
  out  P        annual net premium S A_x / ä_x []
  out  A        whole life insurance A_x []
  out  a_due    whole life annuity-due ä_x []
```

2.5 Solve

Inputs as name=value, each with any unit of the right kind. The answer lists every output, the method used, each input converted to SI, and the conditions the answer assumes.

```
$ actuarial-science solve act_net_premium_whole_life "q=[0.1, 0.2, 0.5, 1]" i=0.05 S=1000.0
P = 306.8418549
A = 0.8656578278
a_due = 2.821185617
  by: Net annual premium for whole life insurance  (equivalence principle: P ä_x = S A_x)
  i = 0.05
  S = 1000
  assumes: the table closes; 0 < i < 1; S > 0
```

2.6 Machine-readable answer

--json prints the same answer as one line of JSON (values in SI): for scripts and other programs.

```
$ actuarial-science solve act_net_premium_whole_life "q=[0.1, 0.2, 0.5, 1]" i=0.05 S=1000.0
--json
{"outcome":"SOLVED","message":"","value":306.841854896,"unit":"","solved_for":"P","si":
[["i",0.05],["S",1000.0]],"values":[["P",306.841854896,""],["A",0.865657827757,""],
["a_due",2.8211856171,""]],"target":"P","phrase":"annual net premium S A_x /
ä_x","assumptions":["the table closes","0 < i < 1","S > 0"],"name":"Net annual premium for
whole life insurance","equation":"equivalence principle: P ä_x = S A_x"}
```

2.7 When an input is missing

No number is guessed: the answer names what is missing, shows a complete call, and the program exits with status 1.

```
$ actuarial-science solve act_net_premium_whole_life "q=[0.1, 0.2, 0.5, 1]" i=0.05
NEEDS_INPUT: act_net_premium_whole_life needs S (sum insured). Example:
solve('act_net_premium_whole_life', q=[0.1, 0.2, 0.5, 1.0], i=0.05, S=1000).
```

2.8 Many questions at once

One JSON request per line in, one JSON answer per line out, in order: for scripts, notebooks and agents (the program starts once). In PowerShell: Get-Content cases.jsonl | actuarial-science batch

`cases.jsonl`:

```
{"key": "act_net_premium_whole_life", "inputs": {"q": "[0.1, 0.2, 0.5, 1]", "i": 0.05, "S":
1000.0}}
{"key": "act_annuity_certain", "inputs": {"n": 10, "i": 0.05}}
```

```
$ actuarial-science batch < cases.jsonl
{"outcome":"SOLVED","message":"","value":306.841854896,"unit":"","solved_for":"P","si":
[["i",0.05],["S",1000.0]],"values":[["P",306.841854896,""],["A",0.865657827757,""],
["a_due",2.8211856171,""]],"target":"P","phrase":"annual net premium S A_x /
ä_x","assumptions":["the table closes","0 < i < 1","S > 0"],"name":"Net annual premium for
whole life insurance","equation":"equivalence principle: P ä_x = S A_x"}
{"outcome":"SOLVED","message":"","value":7.72173492918,"unit":"","solved_for":"a_n","si":
[["i",0.05]],"values":[["a_n",7.72173492918,""],["a_due",8.10782167564,""],
["s_n",12.5778925355,""],["s_due",13.2067871623,""],
["a_cont",7.91320859505,""]],"target":"a_n","phrase":"present value of an annuity-immediate
a_n","assumptions":["n >= 1","0 < i < 1"],"name":"Annuity-certain present and accumulated
values","equation":"a_n = (1 - v^n)/i, a-due = (1 - v^n)/d, s_n = ((1+i)^n - 1)/i, s-due =
((1+i)^n - 1)/d, continuous = (1 - v^n)/delta"}
```

3. Writing inputs

Inputs are written `name=value`, one per argument; `actuarial-science info <capability>` lists the names. Quote an argument that holds a space or brackets (bash, zsh and PowerShell all accept double quotes). The answer always states each input as it was read, converted to SI, so a misread unit is visible at once.

kind	example	how it is read
a number with a unit	`"R=0.128 nm"`	any unit of the same kind; converted to SI
a number in SI	`R=1.28e-10`	no unit means the SI unit shown by info
a choice	`structure=FCC`	one of the values info lists
a list or vector	`"v=[1, 2, 3]" or "v=1, 2, 3"`	brackets keep it a list
a matrix	`"A=[[1, 2], [3, 4]]"`	one list per row
true / false	`inclusive=true`	for yes/no options

An equation can be solved for any of its quantities: give the others and add `--target <symbol>`. Without `--target`, the program solves for the quantity the capability is named after.

4. Outcomes and exit codes

The program never guesses a number: when it cannot answer, it says why and exits with status 1, so a script can tell an answer from a refusal without reading the text.

outcome	meaning	exit status
SOLVED	the answer: every output with its unit, the method, the inputs in SI, the conditions assumed	0
NEEDS_INPUT	an input is missing; the message names it and shows a complete call	1
INVALID_INPUT	a value or unit of the wrong kind, an unknown input name, or a value outside the conditions	1
AMBIGUOUS	the inputs allow two different answers; say which one is meant	1
OUT_OF_SCOPE	the model has no capability for this, or the result would not be meaningful	1
(usage)	a command the program does not know: it prints its usage	2

5. Scripts and agents: batch and JSON

`actuarial-science solve ... --json` prints one answer as one line of JSON. `actuarial-science batch` reads one request per line on standard input and writes one JSON answer per line, in order, starting the program once for any number of requests. A request is `{"key": "<capability>", "inputs": {"name": "value", ...}, "target": null}`. In PowerShell, pipe the file in: `Get-Content cases.jsonl | actuarial-science batch`. `actuarial-science catalog` prints every capability with its inputs as JSON, to build forms or tool definitions.

answer field	meaning
outcome	one of the outcomes above
message	why there is no answer (empty when SOLVED)
value, unit	the main result in SI units
values	every output: [symbol, value, unit] (procedures)
target	the quantity solved for
si	each numeric input converted to SI: [symbol, value]
assumptions	the conditions the answer relies on
name, equation	the capability and its equation or method

```
import json, subprocess

reqs = [{"key": "<capability>", "inputs": {"name": "value"}}]
out = subprocess.run(["actuarial-science", "batch"], input="\n".join(map(json.dumps,
reqs)),
                     capture_output=True, text=True, check=True).stdout
answers = [json.loads(line) for line in out.splitlines()]
```

6. Figures

`actuarial-science figures <capability> name=value ... --out DIR` draws the live figures for those inputs and writes each as an SVG file into DIR (an existing folder); without `--out` the SVG text comes back as JSON. The desktop app and the model page draw the same figures as soon as you press Compute.

7. In a web page or Node.js: WebAssembly

The same model is a WebAssembly module, `actuarial_science_ai999x.wasm` (https://www.ai999x.com/compute/rust/actuarial_science_ai999x.wasm), with a small loader, `ax-wasm.js` (https://www.ai999x.com/compute/rust/ax-wasm.js). It runs in a browser, a Web Worker or Node.js, with no server.

```
<script src="ax-wasm.js"></script>
<script>
  AXWasm.load("actuarial_science_ai999x.wasm").then(function (m) {
    var a = m.solve({key: "<capability>", inputs: {name: "value"}});
    console.log(a.outcome, a.values || a.value);
    console.log(m.catalog().capabilities.length + " capabilities");
  });
</script>
```

8. Every capability

By topic. For each: what it does, what it needs and gives, when it applies, and a command with what the program printed for it. `actuarial-science info <id>` shows the same inputs and outputs in the terminal.

Credibility (7)

Buhlmann credibility premium (known structural parameters), `act_buhlmann_credibility`

The Buhlmann credibility factor Z = n/(n + k), k = EPV/VHM, and the credibility premium Z xbar + (1 - Z) mu from the overall mean, the expected process variance, the variance of the hypothetical means and n observations with mean xbar.

$$k = \frac{\mathrm{EPV}}{\mathrm{VHM}}, \quad Z = \frac{n}{n+k}, \quad P_C = Z\bar{X} + (1-Z)\mu$$

symbol		meaning	SI unit
`mu`	in	overall (collective) mean	,
`EPV`	in	expected process variance	,
`VHM`	in	variance of the hypothetical means	,
`n`	in	number of observations	,
`xbar`	in	mean of the observations	,
`k`	out	Buhlmann k = EPV/VHM	,
`Z`	out	credibility factor n/(n + k)	,
`premium`	out	Buhlmann credibility premium	,

Applies when: EPV, VHM > 0; n > 0.

```
$ actuarial-science solve act_buhlmann_credibility mu=1000 EPV=50000 VHM=10000 n=5
xbar=1200
k = 5
Z = 0.5
premium = 1100
  by: Buhlmann credibility premium (known structural parameters)  (k = EPV/VHM; Z = n/(n +
k); P = Z xbar + (1 - Z) mu)
  mu = 1000
  EPV = 50000
  VHM = 10000
  n = 5
  xbar = 1200
  assumes: EPV, VHM > 0; n > 0
```

What the worked example means: This calculation determines the Buhlmann credibility premium by combining the observed mean with the collective mean using a weight based on data volume and variability. The process begins by computing the structural parameter k as the ratio of the expected process variance to the variance of hypothetical means, yielding 5. Next, the credibility factor Z is calculated using the number of observations n and k, resulting in 0.5. Finally, the premium is obtained by weighting the sample mean xbar by Z and the overall mean mu by one minus Z, giving a final value of 1100. This approach balances individual experience against general population trends.

Buhlmann-Straub credibility with exposures, `act_buhlmann_straub`

The nonparametric Buhlmann-Straub estimates of the expected process variance and of the variance of the hypothetical means from losses per unit of exposure and the exposures of r risks over n periods, each risk's credibility factor and its credibility premium (complement: the exposure-weighted overall mean).

$$\hat{v} = \frac{\sum_{i,j} m_{ij}(X_{ij} - \bar{X}_i)^2}{r(n-1)}, \quad \hat{a} = \frac{\sum_i m_i(\bar{X}_i - \bar{X})^2 - \hat{v}(r-1)}{m - \sum_i m_i^2/m}, \quad Z_i = \frac{m_i}{m_i + \hat{v}/\hat{a}}$$

symbol		meaning	SI unit
`X`	in	losses per unit of exposure: one row per risk, one column per period	,
`m`	in	exposures, the same shape as X	,
`EPV`	out	estimated expected process variance	,
`VHM`	out	estimated variance of the hypothetical means	,
`k`	out	k = EPV/VHM	,
`Z`	out	credibility factor of each risk m_i/(m_i + k) (a list)	,
`premiums`	out	credibility premium per unit of exposure of each risk (a list)	,
`mean`	out	exposure-weighted overall mean	,

Applies when: at least 2 risks and 2 periods; positive exposures; estimated VHM > 0.

```
$ actuarial-science solve act_buhlmann_straub "X=[[100, 120, 110], [200, 180, 230]]" "m=
[[10, 20, 15], [30, 25, 20]]"
EPV = 7661.111111
VHM = 3834.197531
k = 1.998100267
Z = [0.957485509927,0.974050005645]
premiums = [114.590043128,200.466171022]
mean = 167.9166667
  by: Buhlmann-Straub credibility with exposures  (v = sum m_ij (X_ij - Xbar_i)^2 / (r(n-
1)); a = [sum m_i (Xbar_i - Xbar)^2 - v (r-1)] / (m - sum m_i^2/m); Z_i = m_i/(m_i + v/a))
  assumes: at least 2 risks and 2 periods; positive exposures; estimated VHM > 0
```

What the worked example means: This calculation determines the credibility-adjusted premium for each risk based on their loss history and exposure levels. It computes the expected process variance, which reflects random fluctuations within each risk's experience, and the variance of hypothetical means, which captures systematic differences between risks. The resulting credibility factor balances individual experience against the overall portfolio mean. For the current dataset, the estimated process variance is 7661.11 and the variance of hypothetical means is 3834.2, leading to a specific premium value of [114.59, 200.466] per unit of exposure.

Empirical Buhlmann credibility from a data table, `act_buhlmann_empirical`

The nonparametric estimates of the expected process variance and of the variance of the hypothetical means from r risks observed for n periods each, the Buhlmann k and Z, and each risk's credibility premium.

$$\hat{v} = \frac{1}{r(n-1)} \sum_{i,j} \left(X_{ij} - \bar{X}_i \right)^2, \quad \hat{a} = \frac{1}{r-1} \sum_{i} \left(\bar{X}_i - \bar{X} \right)^2 - \frac{\hat{v}}{n}$$

symbol		meaning	SI unit
`X`	in	observations: one row per risk, one column per period (rows of equal length)	,
`EPV`	out	estimated expected process variance	,
`VHM`	out	estimated variance of the hypothetical means	,
`k`	out	Buhlmann k = EPV/VHM	,
`Z`	out	credibility factor n/(n + k)	,
`premiums`	out	credibility premium of each risk (a list)	,
`mean`	out	overall mean	,

Applies when: at least 2 risks and 2 periods; estimated VHM > 0.

```
$ actuarial-science solve act_buhlmann_empirical "X=[[3, 5, 7], [6, 12, 9]]"
EPV = 6.5
VHM = 5.833333333
k = 1.114285714
Z = 0.7291666667
premiums = [5.54166666667,8.45833333333]
mean = 7
  by: Empirical Buhlmann credibility from a data table  (EPV = mean of the within-risk
sample variances; VHM = sample variance of the risk means - EPV/n; Z = n/(n + EPV/VHM))
  assumes: at least 2 risks and 2 periods; estimated VHM > 0
```

What the worked example means: This calculation determines the empirical Buhlmann credibility parameters for a set of risks observed over multiple periods. The expected process variance, estimated at 6.5, measures the average variability within each individual risk's history. The variance of hypothetical means, calculated as 5.83333, captures the spread between the different risks' average outcomes after adjusting for internal noise. Combining these yields the structural parameter k of 1.11429, which leads to a credibility factor Z of 0.729167. This factor dictates how much weight is given to a specific risk's own experience versus the overall portfolio mean of 7 when determining the final premium.

Limited fluctuation full credibility for severity and aggregate losses, `act_limited_fluctuation_severity_aggregate`

The expected number of claims needed for full credibility of the claim severity, of aggregate losses (pure premium) and of the claim frequency, for Poisson claim counts and a severity with the given mean and standard deviation.

$$n_S = n_0\left(1+\frac{\sigma_X^2}{\mu_X^2}\right), \quad n_0 = \left(\frac{z_{(1+p)/2}}{k}\right)^2$$

symbol		meaning	SI unit
`p`	in	probability level (90% or 0.9)	,
`k`	in	tolerance as a fraction of the mean (5% or 0.05)	,
`mean_x`	in	mean claim severity	,
`sd_x`	in	standard deviation of the claim severity	,
`claims_aggregate`	out	claims for full credibility of aggregate losses n0 (1 + CV^2)	,
`claims_severity`	out	claims for full credibility of severity n0 CV^2	,
`claims_frequency`	out	claims for full credibility of frequency n0	,

Applies when: 0 < p < 1; 0 < k < 1; Poisson claim counts.

```
$ actuarial-science solve act_limited_fluctuation_severity_aggregate p=0.9 k=0.05
mean_x=1000 sd_x=2000
claims_aggregate = 5411.086908
claims_severity = 4328.869527
claims_frequency = 1082.217382
  by: Limited fluctuation full credibility for severity and aggregate losses  (n0 =
(z/k)^2; aggregate: n0 (1 + (sd/mean)^2); severity: n0 (sd/mean)^2)
  p = 0.9
  k = 0.05
  mean_x = 1000
  sd_x = 2000
  assumes: 0 < p < 1; 0 < k < 1; Poisson claim counts
```

What the worked example means: This calculation determines the expected number of claims required to achieve full credibility for three distinct loss components. Using a probability level of 0.9 and a tolerance limit of 0.05, this gives the base claim count for frequency. For severity, this base is adjusted by the squared coefficient of variation derived from the mean severity of 1000 and standard deviation of 2000. The result indicates that 1082.22 claims are needed for frequency, 4328.87 for severity, and 5411.09 for aggregate losses.

Limited fluctuation full credibility standard for claim counts, `act_limited_fluctuation_full_credibility`

The expected number of claims needed for full credibility of the claim frequency so that the observed number is within a fraction k of its mean with probability p (Poisson claim counts).

$$n_F = \left(\frac{z_{(1+p)/2}}{k} \right)^2$$

symbol		meaning	SI unit
`p`	in	probability level (90% or 0.9)	,
`k`	in	tolerance as a fraction of the mean (5% or 0.05)	,
`n_full`	out	expected claims needed for full credibility (z/k)^2	,
`z`	out	standard normal quantile z_((1+p)/2)	,

Applies when: 0 < p < 1; 0 < k < 1; Poisson claim counts.

```
$ actuarial-science solve act_limited_fluctuation_full_credibility p=0.9 k=0.05
n_full = 1082.217382
z = 1.644853627
  by: Limited fluctuation full credibility standard for claim counts  (n_F = (z_((1+p)/2) /
k)^2)
  p = 0.9
  k = 0.05
  assumes: 0 < p < 1; 0 < k < 1; Poisson claim counts
```

What the worked example means: This calculation determines the minimum expected number of claims needed to achieve full credibility for claim frequency. It uses a standard normal quantile derived from the desired probability level, which is 1.64485, and divides it by the tolerance fraction 0.05. The resulting value,

1082.22, represents the threshold where there is a 0.9 probability that the observed claim count will fall within 0.05 of its true mean, assuming Poisson distributed counts.

Partial credibility (square root rule), `act_partial_credibility`

The credibility factor Z = min(1, sqrt(n/n_F)) of an experience with n claims against a full credibility standard n_F, and the credibility-weighted estimate Z xbar + (1 - Z) M.

$$Z = \min\left(1, \sqrt{n/n_F}\right), \quad P_C = Z\bar{X} + (1 - Z)M$$

symbol		meaning	SI unit
`n`	in	observed (or expected) number of claims	,
`n_full`	in	full credibility standard in claims	,
`xbar`	in	observed experience (mean or rate)	,
`M`	in	complement of credibility (manual rate)	,
`Z`	out	credibility factor	,
`premium`	out	credibility-weighted estimate	,

Applies when: n >= 0; n_F > 0.

```
$ actuarial-science solve act_partial_credibility n=300 n_full=1082 xbar=800 M=1000
Z = 0.5265589476
premium = 894.6882105
  by: Partial credibility (square root rule)  (Z = min(1, sqrt(n / n_F)); estimate = Z xbar
+ (1 - Z) M)
  n = 300
  n_full = 1082
  xbar = 800
  M = 1000
  assumes: n >= 0; n_F > 0
```

In the worked example, n = 300, n_full = 1082, xbar = 800, M = 1000 go through the equation above and give Z = 0.5265589476, premium = 894.6882105. It applies only when n >= 0; n_F > 0: outside that it says what is wrong instead of returning a number.

Poisson-gamma Bayesian credibility, `act_poisson_gamma_credibility`

The posterior (Bayesian) mean claim frequency of a risk whose Poisson claim rate has a gamma prior (alpha, theta: mean alpha theta), after observing claim counts in n periods, with the equivalent credibility factor and the posterior gamma parameters.

$$E[\Lambda \mid x] = \frac{(\alpha + \sum x_j)\,\theta}{1 + n\theta} = Z\bar{x} + (1 - Z)\alpha\theta, \quad Z = \frac{n\theta}{1 + n\theta}$$

symbol		meaning	SI unit
`alpha`	in	gamma prior shape alpha	,

symbol		meaning	SI unit
`theta`	in	gamma prior scale theta	,
`counts`	in	observed claim counts, one per period (a list)	,
`post_mean`	out	posterior mean claim rate (alpha + sum x) theta/(1 + n theta)	,
`Z`	out	credibility factor n theta/(1 + n theta)	,
`prior_mean`	out	prior mean alpha theta	,
`post_alpha`	out	posterior shape alpha + sum x	,
`post_theta`	out	posterior scale theta/(1 + n theta)	,

Applies when: alpha, theta > 0; whole-number counts.

```
$ actuarial-science solve act_poisson_gamma_credibility alpha=3 theta=0.5 "counts=[2, 1,
3]"
post_mean = 1.8
Z = 0.6
prior_mean = 1.5
post_alpha = 9
post_theta = 0.2
  by: Poisson-gamma Bayesian credibility  (posterior Gamma(alpha + sum x, theta/(1 + n
theta)); its mean = Z xbar + (1 - Z) alpha theta with Z = n theta/(1 + n theta))
  alpha = 3
  theta = 0.5
  assumes: alpha, theta > 0; whole-number counts
```

What the worked example means: This calculation determines the updated expected claim frequency for a risk by combining historical data with prior beliefs. Using a Poisson-gamma Bayesian credibility model, it computes the posterior mean as 1.8. This result is derived from the prior parameters alpha and theta, which establish a baseline expectation of 1.5, and the observed claim counts over n periods. The final estimate is a weighted average where the weight given to the actual experience is represented by the credibility factor Z, calculated here as 0.6. The remaining weight, one minus Z, is assigned to the prior mean, reflecting how much confidence is placed in the new data versus existing assumptions.

Interest theory (11)

Accumulated and present value of a single amount, `act_accumulated_value`

The value of an amount accumulated for t years at compound interest, the present value of the same amount due in t years, and the accumulation factor (1+i)^t.

$$AV = P(1+i)^t, \quad PV = P\,v^t$$

symbol		meaning	SI unit
`P`	in	amount	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`t`	in	time in years	,
`AV`	out	accumulated value P(1+i)^t	,
`PV`	out	present value P v^t	,

symbol		meaning	SI unit
`factor`	out	accumulation factor (1+i)^t	,

Applies when: 0 < i < 1; 0 <= t <= 200.

```
$ actuarial-science solve act_accumulated_value P=1000 i=0.05 t=10
AV = 1628.894627
PV = 613.9132535
factor = 1.628894627
  by: Accumulated and present value of a single amount  (AV = P(1+i)^t; PV = P(1+i)^(-t))
  P = 1000
  i = 0.05
  t = 10
  assumes: 0 < i < 1; 0 <= t <= 200
```

What the worked example means: This calculation determines the future worth of a sum by compounding it over time, yielding an accumulated value of 1628.89. It also computes the current worth of that same future sum, resulting in a present value of 613.913. The accumulation factor used for these conversions is 1.62889, which reflects the growth potential of the annual effective interest rate applied to the initial amount.

Annuity-certain present and accumulated values, `act_annuity_certain`

The present value of n level payments of 1 at the end (immediate) or start (due) of each year, their accumulated values at time n, and the continuously payable annuity over n years.

$$a_{\overline{n}|} = \frac{1 - v^n}{i},\ \ddot{a}_{\overline{n}|} = \frac{1 - v^n}{d},\ s_{\overline{n}|} = \frac{(1+i)^n - 1}{i},\ \bar{a}_{\overline{n}|} = \frac{1 - v^n}{\delta}$$

symbol		meaning	SI unit
`n`	in	number of annual payments (term in years)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`a_n`	out	present value of an annuity-immediate a_n	,
`a_due`	out	present value of an annuity-due	,
`s_n`	out	accumulated value of an annuity-immediate s_n	,
`s_due`	out	accumulated value of an annuity-due	,
`a_cont`	out	present value of a continuous annuity over n years	,

Applies when: n >= 1; 0 < i < 1.

```
$ actuarial-science solve act_annuity_certain n=10 i=0.05
a_n = 7.721734929
a_due = 8.107821676
s_n = 12.57789254
s_due = 13.20678716
a_cont = 7.913208595
  by: Annuity-certain present and accumulated values  (a_n = (1 - v^n)/i, a-due = (1 -
v^n)/d, s_n = ((1+i)^n - 1)/i, s-due = ((1+i)^n - 1)/d, continuous = (1 - v^n)/delta)
  i = 0.05
  assumes: n >= 1; 0 < i < 1
```

What the worked example means: This calculation determines the financial value of a series of equal annual payments over a fixed period. Using an annual effective interest rate of 0.05 and a term of 10 years, the present value of payments made at the end of each year is computed as 7.72173. The corresponding value for payments made at the beginning of each year is 8.10782. Additionally, the future value of these streams at the end of the term is calculated as 12.5779 for immediate payments and 13.2068 for due payments. A continuous payment stream over the same period has a present value of 7.91321. These figures represent the equivalent lump-sum amounts today or at maturity that match the cash flow pattern under the specified interest conditions.

Effective annual rate from a nominal rate, `act_nominal_to_effective_rate`

The annual effective interest and discount rates, force of interest and discount factor equivalent to a nominal interest or discount rate convertible m times a year.

$$1+i=\left(1+\frac{i^{(m)}}{m}\right)^{m}=\left(1-\frac{d^{(m)}}{m}\right)^{-m}$$

symbol		meaning	SI unit
`rate`	in	nominal annual rate (6% or 0.06)	,
`m`	in	conversion periods per year (12 = monthly)	,
`kind`	in	the nominal rate is an interest rate i(m) or a discount rate d(m)	,
`i`	out	annual effective interest rate	,
`d`	out	annual effective discount rate	,
`delta`	out	force of interest	,
`v`	out	annual discount factor	,

Applies when: 0 < rate < 1; 1 <= m <= 365.

```
$ actuarial-science solve act_nominal_to_effective_rate rate=0.06 m=12 kind=interest
i = 0.06167781186
d = 0.05809466033
delta = 0.05985049813
v = 0.9419053397
  by: Effective annual rate from a nominal rate  (1 + i = (1 + i(m)/m)^m or (1 - d(m)/m)^(-
m); d = i/(1+i); delta = ln(1+i))
  rate = 0.06
  kind = interest
  assumes: 0 < rate < 1; 1 <= m <= 365
```

What the worked example means: This calculation converts a nominal rate of 0.06 per year, compounded 12 times annually as an interest or discount type indicated by interest, into equivalent annual measures. The resulting effective annual interest rate is 0.0616778, meaning that one unit of currency grows to this amount after one year under the specified compounding frequency. The corresponding effective annual discount rate is 0.0580947, representing the proportion deducted from the accumulated value to find the present value. The continuous growth rate, known as the force of interest, is calculated as 0.0598505, while the annual discount factor, which converts future values back to the present, is 0.941905.

Effective rates from a constant force of interest, `act_force_of_interest`

The annual effective interest rate, discount rate and discount factor for a constant force of interest delta.

$$i = e^{\delta} - 1, \quad d = 1 - e^{-\delta}, \quad v = e^{-\delta}$$

symbol		meaning	SI unit
`delta`	in	constant force of interest per year (0.05 or 5%)	,
`i`	out	annual effective interest rate	,
`d`	out	annual effective discount rate	,
`v`	out	annual discount factor	,

Applies when: 0 < delta < 1.

```
$ actuarial-science solve act_force_of_interest delta=0.05
i = 0.05127109638
d = 0.0487705755
v = 0.9512294245
  by: Effective rates from a constant force of interest  (i = e^delta - 1, d = 1 - e^(-
delta), v = e^(-delta))
  delta = 0.05
  assumes: 0 < delta < 1
```

What the worked example means: This calculation determines the annual effective interest rate, discount rate, and discount factor derived from a constant force of interest. Given a force of interest of 0.05, the resulting annual effective interest rate is 0.0512711, the annual effective discount rate is 0.0487706, and the annual discount factor is 0.951229. These values represent the equivalent yearly rates that produce the same accumulation as the continuous compounding defined by the input force.

Equivalent rates from an effective annual interest rate, `act_interest_rate_conversion`

The effective discount rate, discount factor, force of interest and the nominal interest and discount rates convertible m times a year equivalent to an annual effective interest rate.

$$d = \frac{i}{1+i}, \; \delta = \ln(1+i), \; i^{(m)} = m\left[(1+i)^{1/m} - 1\right], \; d^{(m)} = m\left[1 - v^{1/m}\right]$$

symbol		meaning	SI unit
`i`	in	annual effective interest rate (5% or 0.05)	,
`m`	in	conversion periods per year for the nominal rates (12 = monthly)	,
`d`	out	effective annual discount rate	,
`v`	out	annual discount factor 1/(1+i)	,
`delta`	out	force of interest ln(1+i)	,
`i_m`	out	nominal interest rate convertible m-thly	,
`d_m`	out	nominal discount rate convertible m-thly	,

Applies when: 0 < i < 1; 1 <= m <= 365.

```
$ actuarial-science solve act_interest_rate_conversion i=0.05 m=12
d = 0.04761904762
v = 0.9523809524
delta = 0.04879016417
i_m = 0.0488894854
d_m = 0.04869111179
  by: Equivalent rates from an effective annual interest rate  (d = i/(1+i), v = 1/(1+i),
delta = ln(1+i), i(m) = m[(1+i)^(1/m) - 1], d(m) = m[1 - v^(1/m)])
  i = 0.05
  assumes: 0 < i < 1; 1 <= m <= 365
```

What the worked example means: This calculation derives several key financial rates from a single annual effective interest rate, 0.05. It computes the effective annual discount rate, 0.047619, which represents the proportion of the accumulated amount that constitutes interest. The annual discount factor, 0.952381, indicates the present value of one unit received in one year. Additionally, it determines the force of interest, 0.0487902, representing continuous compounding equivalence. Finally, nominal rates convertible m times per year are calculated: the nominal interest rate, 0.0488895, and the nominal discount rate, 0.0486911, using the specified conversion frequency, 12. These values ensure consistency across different compounding conventions.

Increasing and decreasing annuities-certain, `act_increasing_decreasing_annuity`

The present values of an n-year annuity paying 1, 2, ..., n (increasing) and n, n-1, ..., 1 (decreasing), immediate and due.

$$(Ia)_{\overline{n}|} = \frac{\ddot{a}_{\overline{n}|} - nv^n}{i}, \quad (Da)_{\overline{n}|} = \frac{n - a_{\overline{n}|}}{i}$$

symbol		meaning	SI unit
`n`	in	term in years	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`Ia`	out	increasing annuity-immediate (Ia)_n	,
`Ia_due`	out	increasing annuity-due	,
`Da`	out	decreasing annuity-immediate (Da)_n	,
`Da_due`	out	decreasing annuity-due	,

Applies when: n >= 1; 0 < i < 1.

```
$ actuarial-science solve act_increasing_decreasing_annuity n=10 i=0.05
Ia = 39.3737828
Ia_due = 41.34247194
Da = 45.56530142
Da_due = 47.84356649
  by: Increasing and decreasing annuities-certain  ((Ia)_n = (a-due_n - n v^n)/i; (Da)_n =
(n - a_n)/i; due values times (1+i))
  i = 0.05
  assumes: n >= 1; 0 < i < 1
```

What the worked example means: This calculation determines the present values of linearly changing annuities certain over a term of 10 years at an annual effective interest rate of 0.05. The results include the increasing annuity immediate, which pays one unit at the end of each year rising to n units, valued at 39.3738, and the decreasing annuity immediate, which starts at n units and falls to one, valued at 45.5653. Corresponding annuity-due values, where payments occur at the beginning of each period, are calculated as 41.3425 for the increasing case and 47.8436 for the decreasing case.

Loan amortization: level payment, interest and principal, outstanding balance, `act_loan_amortization`

The level annual payment that repays a loan over n years, the interest and principal parts of payment t and the outstanding balance just after it.

$$R = \frac{L}{a_{\overline{n}|}}, \quad I_t = R\left(1 - v^{n-t+1}\right), \quad P_t = R\,v^{n-t+1}, \quad B_t = R\,a_{\overline{n-t}|}$$

symbol		meaning	SI unit
`L`	in	loan amount	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`n`	in	number of annual payments	,
`t`	in	payment number of interest (1 to n)	,
`R`	out	level annual payment	,
`interest_t`	out	interest part of payment t	,
`principal_t`	out	principal repaid in payment t	,
`balance_t`	out	outstanding balance just after payment t	,
`total_interest`	out	total interest paid over the loan	,

Applies when: L > 0; 0 < i < 1; 1 <= t <= n.

```
$ actuarial-science solve act_loan_amortization L=100000 i=0.06 n=20 t=5
R = 8718.455698
interest_t = 5286.468013
principal_t = 3431.987685
balance_t = 84675.81252
total_interest = 74369.11395
  by: Loan amortization: level payment, interest and principal, outstanding balance  (R =
L/a_n; balance before payment t = R a_(n-t+1); interest = i x balance; principal = R v^(n-
t+1); balance after = R a_(n-t))
  L = 100000
  i = 0.06
  assumes: L > 0; 0 < i < 1; 1 <= t <= n
```

What the worked example means: This calculation determines the fixed annual payment required to repay a loan of amount 100000 over 20 years at an effective interest rate of 0.06. The resulting level payment is 8718.46. For the specific payment number 5, the interest component is 5286.47 and the principal repayment is 3431.99, leaving an outstanding balance just after this payment of 84675.8. The total interest paid over the entire life of the loan is 74369.1. This method assumes payments are made at the end of each period and that the interest rate remains constant throughout the term.

Macaulay and modified duration and convexity of cash flows, `act_duration_convexity`

The present value, Macaulay duration, modified duration and convexity (second derivative of price with respect to i over price) of a set of non-negative cash flows.

$$D_{mac} = \frac{\sum t\, C_t v^t}{P}, \quad D_{mod} = \frac{D_{mac}}{1+i}, \quad C = \frac{\sum t(t+1) C_t v^{t+2}}{P}$$

symbol		meaning	SI unit
`amounts`	in	cash flow amounts (a list, non-negative)	,
`times`	in	payment times in years (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`PV`	out	present value	,
`macaulay`	out	Macaulay duration (years)	,
`modified`	out	modified duration -P'(i)/P	,
`convexity`	out	convexity P''(i)/P	,

Applies when: amounts >= 0 with a positive total; one time per amount; 0 < i < 1.

```
$ actuarial-science solve act_duration_convexity "amounts=[5, 5, 105]" "times=[1, 2, 3]"
i=0.05
PV = 100
macaulay = 2.859410431
modified = 2.723248029
convexity = 10.2056242
  by: Macaulay and modified duration and convexity of cash flows  (D_mac = sum t C v^t / P;
D_mod = D_mac/(1+i); convexity = sum t(t+1) C v^(t+2) / P)
  i = 0.05
  assumes: amounts >= 0 with a positive total; one time per amount; 0 < i < 1
```

What the worked example means: This calculation determines the price sensitivity of a cash flow stream to changes in interest rates. Using an annual effective interest rate of 0.05, the present value of the payments is computed as 100. The Macaulay duration, representing the weighted average time until cash flows are received, is found to be 2.85941 years. From this, the modified duration, which estimates the percentage change in price for a unit change in yield, is derived as 2.72325. Finally, the convexity measure, capturing the curvature of the price-yield relationship, is calculated as 10.2056. These metrics collectively indicate how much the asset's value will fluctuate given shifts in market rates.

Present value of a perpetuity, `act_perpetuity`

The present value of payments of 1 a year forever, at the end (immediate) or start (due) of each year, and paid continuously.

$$a_{\overline{\infty}|} = \frac{1}{i}, \quad \ddot{a}_{\overline{\infty}|} = \frac{1}{d}, \quad \bar{a}_{\overline{\infty}|} = \frac{1}{\delta}$$

symbol		meaning	SI unit
`i`	in	annual effective interest rate (5% or 0.05)	,
`a_inf`	out	perpetuity-immediate 1/i	,
`a_due_inf`	out	perpetuity-due 1/d	,
`a_cont_inf`	out	continuous perpetuity 1/delta	,

Applies when: 0 < i < 1.

```
$ actuarial-science solve act_perpetuity i=0.05
a_inf = 20
a_due_inf = 21
a_cont_inf = 20.49593431
  by: Present value of a perpetuity  (a_inf = 1/i, a-due_inf = 1/d = (1+i)/i, continuous
1/delta)
  i = 0.05
  assumes: 0 < i < 1
```

What the worked example means: This calculation determines the current worth of an infinite series of payments. Given an annual effective interest rate of 0.05, the value of one unit paid at the end of each year forever is 20. If payments occur at the start of each year, the value rises to 21. For payments made continuously throughout the year, the equivalent present value is 20.4959. These figures represent the lump sum required today to fund these perpetual cash flows under the specified interest conditions.

Present value of a series of cash flows, `act_present_value_cash_flows`

The present value at time 0 of amounts paid at stated times (years), and their accumulated value at the last payment time.

$$PV = \sum_{k} C_k \, v^{t_k}$$

symbol		meaning	SI unit
`amounts`	in	cash flow amounts (a list)	,
`times`	in	payment times in years (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`PV`	out	present value at time 0	,
`AV_at_last`	out	accumulated value at the last payment time	,

Applies when: one time per amount; 0 <= t <= 200; 0 < i < 1.

```
$ actuarial-science solve act_present_value_cash_flows "amounts=[100, 200, 300]" "times=[1,
2, 3]" i=0.05
PV = 535.7952705
AV_at_last = 620.25
  by: Present value of a series of cash flows  (PV = sum C_k v^(t_k); AV = PV
(1+i)^(t_max))
  i = 0.05
  assumes: one time per amount; 0 <= t <= 200; 0 < i < 1
```

What the worked example means: This calculation determines the present value of a series of cash flows by discounting each amount back to time zero using the annual effective interest rate. The result, 535.795, represents what those future payments are worth today. Additionally, the model computes the accumulated value at the time of the final payment, shown as 620.25, which reflects how much the total would grow if invested until that date.

Yield rate of a level annuity, `act_annuity_yield`

The annual effective interest rate at which n level payments R at the ends of the years are worth a given present value.

$$R\,\frac{1-(1+i)^{-n}}{i} = PV$$

symbol		meaning	SI unit
`PV`	in	present value (price) of the annuity	,
`R`	in	level annual payment	,
`n`	in	number of annual payments	,
`i`	out	yield rate (annual effective)	,
`a_n`	out	annuity factor PV/R	,

Applies when: 0 < PV < n R; yield below 99%.

```
$ actuarial-science solve act_annuity_yield PV=7721.73 R=1000 n=10
i = 0.05000013145
a_n = 7.72173
  by: Yield rate of a level annuity  (solve R a_n(i) = PV for i by bisection (a_n decreases
in i, so the root is unique))
  PV = 7721.73
  R = 1000
  assumes: 0 < PV < n R; yield below 99%
```

What the worked example means: This calculation determines the annual effective interest rate, or yield, at which a series of level payments equals the given present value. By solving for the rate where the annuity factor matches the ratio of price to payment, the yield is 0.0500001. This result implies that if you invest 7721.73 today at this rate, it will generate exactly enough funds to cover the scheduled payments of 1000 over 10 years.

Life annuities (5)

Deferred whole life annuity-due, `act_deferred_annuity`

The actuarial present value of a whole life annuity-due of P a year starting u years from now if the life is then alive (u|ä_x = uE_x ä_(x+u)).

$${}_{u|}\ddot{a}_x = \sum_{k\ge u} v^k\,{}_kp_x = {}_uE_x\,\ddot{a}_{x+u}$$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... ending with 1 (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`u`	in	deferment in years	,
`P`	in	annual payment	,
`ua_due`	out	deferred annuity-due u|ä_x	,
`EPV`	out	actuarial present value P u|ä_x	,

Applies when: the table closes; u < length of the table; 0 < i < 1.

```
$ actuarial-science solve act_deferred_annuity "q=[0.1, 0.2, 0.5, 1]" i=0.05 u=2 P=1000
ua_due = 0.96404276
EPV = 964.04276
  by: Deferred whole life annuity-due  (u|ä_x = sum_(k>=u) v^k k_p_x = u_E_x ä_(x+u))
  i = 0.05
  P = 1000
  assumes: the table closes; u < length of the table; 0 < i < 1
```

What the worked example means: This calculation determines the current worth of a series of annual payments that begin after a specified deferment period, provided the individual survives to that time. The result depends on the death probabilities, the interest rate, and the payment amount. For an input payment of 1000 and a deferment of 2 years at an interest rate of 0.05, the resulting deferred annuity-due factor is 0.964043. Multiplying this factor by the payment amount yields the total expected present value of 964.043. This figure represents the lump sum today that would be equivalent to the future stream of conditional payments.

m-thly whole life annuity-due under UDD, `act_mthly_annuity_udd`

The actuarial present value of a whole life annuity-due paying 1/m at the start of every 1/m of a year, from an annual life table under the uniform distribution of deaths (ä^(m)_x = alpha(m) ä_x - beta(m)).

$$\ddot{a}_x^{(m)} = \alpha(m)\,\ddot{a}_x - \beta(m), \quad \alpha(m) = \frac{id}{i^{(m)}d^{(m)}},\ \beta(m) = \frac{i - i^{(m)}}{i^{(m)}d^{(m)}}$$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... ending with 1 (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`m`	in	payments per year (12 = monthly)	,
`a_m`	out	m-thly annuity-due ä^(m)_x (total 1 a year)	,
`alpha`	out	alpha(m) = i d/(i(m) d(m))	,
`beta`	out	beta(m) = (i - i(m))/(i(m) d(m))	,
`a_due`	out	annual annuity-due ä_x	,

Applies when: the table closes; 0 < i < 1; 1 <= m <= 365; UDD within each year.

```
$ actuarial-science solve act_mthly_annuity_udd "q=[0.1, 0.2, 0.5, 1]" i=0.05 m=12
a_m = 2.355233403
alpha = 1.000197011
beta = 0.4665080196
a_due = 2.821185617
  by: m-thly whole life annuity-due under UDD  (ä^(m)_x = alpha(m) ä_x - beta(m))
  i = 0.05
  assumes: the table closes; 0 < i < 1; 1 <= m <= 365; UDD within each year
```

What the worked example means: This calculation determines the actuarial present value of a whole life annuity-due with m payments per year. Starting with an annual effective interest rate of 0.05 and death probabilities, this gives the annual annuity-due value, which is 2.82119. The uniform distribution of deaths assumption then adjusts this annual figure into the more frequent payment schedule. The adjustment relies on two coefficients: alpha(m), calculated as 1.0002, and beta(m), calculated as 0.466508. The final result, 2.35523, represents the total expected present value of receiving one unit per year in m equal installments at the start of each period.

Temporary life annuity, `act_temporary_annuity`

The actuarial present values of an n-year temporary life annuity-due (ä_(x:n)) and annuity-immediate (a_(x:n)), and the value of an annual payment P paid in advance.

$$\ddot{a}_{x:\overline{n}|} = \sum_{k=0}^{n-1} v^k \, {}_kp_x, \quad a_{x:\overline{n}|} = \ddot{a}_{x:\overline{n}|} - 1 + {}_nE_x$$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... for at least n years (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`n`	in	number of years	,
`P`	in	annual payment	,
`a_due`	out	temporary annuity-due ä_(x:n)	,
`a_imm`	out	temporary annuity-immediate a_(x:n)	,
`EPV`	out	actuarial present value P ä_(x:n)	,

Applies when: at least n death probabilities; 0 < i < 1; P > 0.

```
$ actuarial-science solve act_temporary_annuity "q=[0.1, 0.2, 0.5, 1]" i=0.05 n=3 P=1000
a_due = 2.510204082
a_imm = 1.821185617
EPV = 2510.204082
  by: Temporary life annuity  (ä_(x:n) = sum_(k<n) v^k k_p_x; a_(x:n) = ä_(x:n) - 1 +
n_E_x)
  i = 0.05
  P = 1000
  assumes: at least n death probabilities; 0 < i < 1; P > 0
```

What the worked example means: This calculation determines the present value of an n-year temporary life annuity. It sums the discounted survival probabilities to find the annuity-due factor, which represents the

expected value of one unit paid annually in advance while the individual survives. The result shows that with an interest rate of 0.05 and a term of 3 years, the annuity-due value is 2.5102. Consequently, the total actuarial present value for the specified annual payment P is 2510.2. This figure helps insurers price premiums or reserves by quantifying the current worth of future contingent payments.

Whole life annuity, `act_whole_life_annuity`

The actuarial present values of a whole life annuity-due (ä_x) and annuity-immediate (a_x) for a life aged x, the value of an annual payment P, and the standard deviation of the present value of the annuity-due payments.

$$\ddot{a}_x = \sum_{k \geq 0} v^k \, {}_kp_x, \quad a_x = \ddot{a}_x - 1, \quad \operatorname{Var}(Y) = \frac{{}^2A_x - A_x^2}{d^2}$$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... ending with 1 (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`P`	in	annual payment	,
`a_due`	out	whole life annuity-due ä_x	,
`a_imm`	out	whole life annuity-immediate a_x	,
`EPV`	out	actuarial present value P ä_x	,
`sd`	out	standard deviation of the annuity-due present value	,

Applies when: the table closes; 0 < i < 1; P > 0.

```
$ actuarial-science solve act_whole_life_annuity "q=[0.1, 0.2, 0.5, 1]" i=0.05 P=1000
a_due = 2.821185617
a_imm = 1.821185617
EPV = 2821.185617
sd = 872.3551839
  by: Whole life annuity  (ä_x = sum_k v^k k_p_x; a_x = ä_x - 1; Var = P^2 (2A_x -
A_x^2)/d^2)
  i = 0.05
  P = 1000
  assumes: the table closes; 0 < i < 1; P > 0
```

What the worked example means: This calculation determines the actuarial present values for a whole life annuity. The annuity-due value, representing payments made at the start of each year while alive, is computed as 2.82119. Consequently, the annuity-immediate value, where payments occur at the end of the year, is derived by subtracting one from the due value, resulting in 1.82119. Multiplying the annuity-due value by the annual payment amount 1000 gives the total expected present value of the income stream, which is 2821.19. This figure represents the lump sum equivalent today of receiving 1000 annually for the remainder of the life's duration, discounted for both time value and survival probability.

Woolhouse approximation for an m-thly annuity, `act_woolhouse_approximation`

Woolhouse's two- and three-term approximations of the m-thly whole life annuity-due ä^(m)_x from the annual annuity-due ä_x, the interest rate and the force of mortality at x.

$$\ddot{a}_x^{(m)} \approx \ddot{a}_x - \frac{m-1}{2m} - \frac{m^2-1}{12m^2}(\delta + \mu_x)$$

symbol		meaning	SI unit
`a_due`	in	annual whole life annuity-due ä_x	,
`m`	in	payments per year (12 = monthly)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`mu`	in	force of mortality at age x (per year)	,
`a_m3`	out	three-term Woolhouse approximation	,
`a_m2`	out	two-term approximation ä_x - (m-1)/(2m)	,

Applies when: ä_x >= 1; 1 <= m <= 365; 0 < i < 1; 0 <= mu < 5.

```
$ actuarial-science solve act_woolhouse_approximation a_due=14.8166 m=12 i=0.05 mu=0.00473
a_m3 = 14.35383763
a_m2 = 14.35826667
  by: Woolhouse approximation for an m-thly annuity  (ä^(m)_x ≈ ä_x - (m-1)/(2m) - (m^2-
1)/(12 m^2) (delta + mu_x))
  a_due = 14.8166
  i = 0.05
  mu = 0.00473
  assumes: ä_x >= 1; 1 <= m <= 365; 0 < i < 1; 0 <= mu < 5
```

What the worked example means: This calculation estimates the present value of an annuity-due with payments made m times per year. It starts with the annual annuity-due value, 14.8166, and adjusts it downward to account for more frequent payments. The two-term result, 14.3583, subtracts a fixed adjustment based on payment frequency. The three-term result, 14.3538, includes an additional correction that depends on the interest rate, 0.05, and the force of mortality, 0.00473, providing a more precise valuation when these factors are significant.

Life insurance (5)

Continuous whole life insurance and annuity under UDD, `act_continuous_insurance_udd`

The actuarial present value of a whole life insurance paid at the moment of death (Ā_x = (i/delta) A_x) and of a continuous whole life annuity, from an annual life table with deaths uniformly distributed within each year.

$$\bar{A}_x = \frac{i}{\delta} A_x, \qquad \bar{a}_x = \frac{1 - \bar{A}_x}{\delta}$$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... ending with 1 (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`A_bar`	out	insurance payable at the moment of death Ā_x	,
`A`	out	insurance payable at the end of the year of death A_x	,
`a_bar`	out	continuous whole life annuity ā_x	,

Applies when: the table closes; 0 < i < 1; uniform distribution of deaths within each year.

```
$ actuarial-science solve act_continuous_insurance_udd "q=[0.1, 0.2, 0.5, 1]" i=0.05
A_bar = 0.8871232988
A = 0.8656578278
a_bar = 2.313513453
  by: Continuous whole life insurance and annuity under UDD  (Ā_x = (i/delta) A_x; ā_x = (1
- Ā_x)/delta)
  i = 0.05
  assumes: the table closes; 0 < i < 1; uniform distribution of deaths within each year
```

What the worked example means: This calculation determines the actuarial present values for a whole life insurance policy that pays at the moment of death and a continuous whole life annuity. It uses an annual life table with uniform distribution of deaths within each year to convert discrete end-of-year payments into continuous ones. The resulting insurance value is 0.887123, while the corresponding annuity value is 2.31351. These figures represent the current cost or benefit value based on the input interest rate of 0.05.

Deferred whole life insurance, `act_deferred_insurance`

The actuarial present value of a whole life insurance of S that pays only if death occurs after u years (u|A_x = uE_x A_(x+u)).

$$ {}_{u|}A_x = \sum_{k \geq u} v^{k+1}\, {}_kp_x\, q_{x+k} = {}_uE_x\, A_{x+u} $$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... ending with 1 (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`u`	in	deferment in years	,
`S`	in	sum insured	,
`uA`	out	deferred insurance u|A_x per unit	,
`EPV`	out	actuarial present value S u|A_x	,
`uEx`	out	pure endowment u_E_x	,

Applies when: the table closes; u < length of the table; 0 < i < 1.

```
$ actuarial-science solve act_deferred_insurance "q=[0.1, 0.2, 0.5, 1]" i=0.05 u=1 S=1000
uA = 0.7704197325
EPV = 770.4197325
uEx = 0.8571428571
  by: Deferred whole life insurance  (u|A_x = sum_(k>=u) v^(k+1) k_p_x q_(x+k) = u_E_x
A_(x+u))
  i = 0.05
  S = 1000
  assumes: the table closes; u < length of the table; 0 < i < 1
```

What the worked example means: This calculation determines the present value of a whole life insurance policy that only pays out if death occurs after a specified deferment period. It uses the input sum insured of 1000 and interest rate of 0.05 to compute the unit deferred insurance value 0.77042 and the pure endowment

factor 0.857143. The final result, 770.42, represents the current financial liability for this contingent payment, reflecting both mortality risks and time value of money.

Endowment insurance (fully discrete), `act_endowment_insurance`

The actuarial present value of an n-year endowment insurance paying S at the end of the year of death within n years or at time n on survival, split into its term insurance and pure endowment parts, with the second moment and standard deviation.

$$A_{x:\overline{n}|} = A^{1}_{x:\overline{n}|} + {}_{n}E_{x}$$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... for at least n years (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`n`	in	term in years	,
`S`	in	sum insured	,
`A`	out	endowment insurance A_(x:n) per unit	,
`A1`	out	term insurance part A^1_(x:n)	,
`nEx`	out	pure endowment part n_E_x	,
`A2`	out	second moment at double the force of interest	,
`EPV`	out	actuarial present value S A_(x:n)	,
`sd`	out	standard deviation of the present value	,

Applies when: at least n death probabilities; 0 < i < 1; S > 0.

```
$ actuarial-science solve act_endowment_insurance "q=[0.1, 0.2, 0.5, 1]" i=0.05 n=2 S=1000
A = 0.9115646258
A1 = 0.2585034014
nEx = 0.6530612245
A2 = 0.8311351752
EPV = 911.5646258
sd = 13.60544218
  by: Endowment insurance (fully discrete)  (A_(x:n) = A^1_(x:n) + n_E_x)
  i = 0.05
  S = 1000
  assumes: at least n death probabilities; 0 < i < 1; S > 0
```

What the worked example means: This calculation determines the expected cost of an n-year endowment insurance policy. It combines the term insurance component, valued at 0.258503 per unit, and the pure endowment component, valued at 0.653061 per unit, to find the total single premium factor of 0.911565. Multiplying this factor by the sum insured yields the final actuarial present value of 911.565. The result reflects the time value of money at interest rate 0.05 and mortality assumptions over the term.

Term insurance (fully discrete), `act_term_insurance`

The actuarial present value of an n-year term insurance paying S at the end of the year of death if death occurs within n years, its second moment and the standard deviation of the present value.

$$A^{1}_{x:\overline{n}|} = \sum_{k=0}^{n-1} v^{k+1}\ {}_{k}p_{x}\ q_{x+k}$$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... for at least n years (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`n`	in	term in years	,
`S`	in	sum insured	,
`A1`	out	term insurance A^1_(x:n) per unit	,
`A1_2`	out	second moment at double the force of interest	,
`EPV`	out	actuarial present value S A^1_(x:n)	,
`sd`	out	standard deviation of the present value	,

Applies when: at least n death probabilities; 0 < i < 1; S > 0.

```
$ actuarial-science solve act_term_insurance "q=[0.1, 0.2, 0.5, 1]" i=0.05 n=2 S=1000
A1 = 0.2585034014
A1_2 = 0.2387893933
EPV = 258.5034014
sd = 414.6870926
  by: Term insurance (fully discrete)  (A^1_(x:n) = sum_(k<n) v^(k+1) k_p_x q_(x+k))
  i = 0.05
  S = 1000
  assumes: at least n death probabilities; 0 < i < 1; S > 0
```

What the worked example means: This calculation determines the expected cost of an n-year term insurance policy. It sums the discounted probabilities of death for each year in the term, multiplied by the sum insured. The result, 258.503, represents the fair premium per unit of coverage before loading, while 414.687 indicates the volatility of the claim payout.

Whole life insurance (fully discrete), `act_whole_life_insurance`

The actuarial present value of a whole life insurance paying S at the end of the year of death of a life aged x (A_x), its second moment at double the force of interest, and the standard deviation of the present value of the benefit.

$$A_x = \sum_{k\geq 0} v^{k+1}\ {}_{k}p_{x}\ q_{x+k}, \quad \operatorname{Var}(Z) = S^2\left({}^{2}A_x - A_x^2\right)$$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... ending with 1 (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`S`	in	sum insured	,
`A`	out	A_x per unit sum insured	,
`A2`	out	second moment 2A_x (at double the force of interest)	,

symbol		meaning	SI unit
`EPV`	out	actuarial present value S A_x	,
`sd`	out	standard deviation of the present value S sqrt(2A - A^2)	,

Applies when: the table closes (last q = 1); 0 < i < 1; S > 0.

```
$ actuarial-science solve act_whole_life_insurance "q=[0.1, 0.2, 0.5, 1]" i=0.05 S=1000
A = 0.8656578278
A2 = 0.7510891064
EPV = 865.6578278
sd = 41.54072304
  by: Whole life insurance (fully discrete)  (A_x = sum_k v^(k+1) k_p_x q_(x+k); 2A_x with
v^2; Var Z = S^2 (2A - A^2))
  i = 0.05
  S = 1000
  assumes: the table closes (last q = 1); 0 < i < 1; S > 0
```

What the worked example means: This calculation determines the expected present value of a whole life insurance policy that pays a benefit at the end of the year of death. Using the input interest rate 0.05 and sum insured 1000, it computes the net single premium factor 0.865658 and its second moment 0.751089. The final result, 865.658, represents the insurer's current liability for this specific contract before accounting for expenses or profit margins.

Life tables (9)

Commutation functions D, N, C and M, `act_commutation_functions`

The commutation functions D_x = v^x l_x, N_x, C_x = v^(x+1) d_x and M_x at age x of a complete life table, with the whole life insurance A_x = M_x/D_x and annuity-due ä_x = N_x/D_x.

$$D_x = v^x l_x,\ C_x = v^{x+1} d_x,\ N_x = \sum_{y \geq x} D_y,\ M_x = \sum_{y \geq x} C_y,\ A_x = \frac{M_x}{D_x},\ \ddot{a}_x = \frac{N_x}{D_x}$$

symbol		meaning	SI unit
`lx`	in	survivors at consecutive integer ages from x0, ending with 0 (a list)	,
`x0`	in	age of the first entry of lx	,
`x`	in	age at which the functions are wanted	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`D_x`	out	D_x = v^x l_x	,
`N_x`	out	N_x = sum of D_y for y >= x	,
`C_x`	out	C_x = v^(x+1) d_x	,
`M_x`	out	M_x = sum of C_y for y >= x	,
`A_x`	out	whole life insurance M_x/D_x	,
`a_due_x`	out	whole life annuity-due N_x/D_x	,

Applies when: the table closes (last lx = 0); x inside the table; 0 < i < 1.

```
$ actuarial-science solve act_commutation_functions "lx=[1000, 900, 700, 400, 0]" x0=95
x=95 i=0.05
D_x = 9.705470376
N_x = 27.54022849
C_x = 0.924330512
M_x = 8.394030925
A_x = 0.8648762604
a_due_x = 2.837598531
  by: Commutation functions D, N, C and M  (D_y = v^y l_y, C_y = v^(y+1) d_y, N_x = sum
D_y, M_x = sum C_y; A_x = M_x/D_x, ä_x = N_x/D_x)
  i = 0.05
  assumes: the table closes (last lx = 0); x inside the table; 0 < i < 1
```

In the worked example, lx = [1000, 900, 700, 400, 0], x0 = 95, x = 95, i = 0.05 go through the equation above and give D_x = 9.705470376, N_x = 27.54022849, C_x = 0.924330512, M_x = 8.394030925. It applies only when the table closes (last lx = 0); x inside the table; 0 < i < 1: outside that it says what is wrong instead of returning a number.

Curtate and complete expectation of life, `act_life_expectancy`

The curtate expectation of life e_x (whole years lived) and the complete expectation e°_x under a uniform distribution of deaths within each year, for a life aged x in a complete life table.

$$e_x = \sum_{k \geq 1} {}_kp_x, \quad \mathring{e}_x = e_x + \tfrac{1}{2}$$

symbol		meaning	SI unit
`lx`	in	survivors at consecutive integer ages from x0, ending with 0 (a list)	,
`x0`	in	age of the first entry of lx	,
`x`	in	age of the life	,
`e_curtate`	out	curtate expectation of life e_x (years)	,
`e_complete`	out	complete expectation of life under UDD (years)	,

Applies when: the table closes (last lx = 0); x inside the table.

```
$ actuarial-science solve act_life_expectancy "lx=[1000, 900, 700, 400, 0]" x0=95 x=95
e_curtate = 2
e_complete = 2.5
  by: Curtate and complete expectation of life  (e_x = sum_(k>=1) l_(x+k)/l_x; e°_x = e_x +
1/2 under UDD)
  assumes: the table closes (last lx = 0); x inside the table
```

What the worked example means: This calculation determines the expected future lifetime for a person aged 95. The curtate expectation, 2, represents the average number of whole years lived beyond age 95, derived from the survival probabilities in the provided life table. The complete expectation, 2.5, adjusts this figure by adding half a year to account for deaths occurring uniformly throughout each year, providing a more precise estimate of total time lived.

Death probability from the central death rate, `act_central_death_rate`

The one-year death probability q_x corresponding to a central death rate m_x (deaths per year lived), under a uniform distribution of deaths or a constant force of mortality within the year.

$$\text{UDD: } q_x = \frac{2m_x}{2 + m_x}, \qquad \text{constant force: } q_x = 1 - e^{-m_x}$$

symbol		meaning	SI unit
`m`	in	central death rate m_x (deaths per person-year)	,
`assumption`	in	fractional-age assumption: udd or constant_force	,
`q`	out	one-year death probability q_x	,
`p`	out	one-year survival probability p_x	,

Applies when: m > 0; m < 2 under UDD.

```
$ actuarial-science solve act_central_death_rate m=0.02 assumption=udd
q = 0.0198019802
p = 0.9801980198
  by: Death probability from the central death rate  (UDD: q = 2m/(2+m); constant force: q
= 1 - e^(-m))
  m = 0.02
  assumption = udd
  assumes: m > 0; m < 2 under UDD
```

What the worked example means: This calculation determines the likelihood of dying within a single year, denoted as the death probability, which equals 0.019802. It also provides the complementary survival probability, which is 0.980198. These values are computed using the central death rate of 0.02, representing deaths per person-year lived. The result reflects how many individuals are expected to die versus survive over the period, based on the specified mortality assumption.

Deferred death probability u|t q x, `act_deferred_death_probability`

The probability that a life aged x survives u years and then dies within the following t years, from survivor numbers lx starting at age x0.

$$ {}_{u|t}q_x = \frac{l_{x+u} - l_{x+u+t}}{l_x} = {}_{u}p_x \; {}_{t}q_{x+u} $$

symbol		meaning	SI unit
`lx`	in	survivors at consecutive integer ages from x0 (a list)	,
`x0`	in	age of the first entry of lx	,
`x`	in	age of the life	,
`u`	in	years of deferment	,
`t`	in	length of the death interval in years	,
`prob`	out	deferred death probability u|t q_x	,
`upx`	out	probability of surviving the deferment	,

Applies when: x + u + t inside the table; t >= 1.

```
$ actuarial-science solve act_deferred_death_probability "lx=[1000, 950, 880, 780, 650]"
x0=60 x=60 u=2 t=2
prob = 0.23
upx = 0.88
  by: Deferred death probability u|t q x  (u|t q_x = (l_(x+u) - l_(x+u+t))/l_x = u_p_x *
t_q_(x+u))
  assumes: x + u + t inside the table; t >= 1
```

What the worked example means: This calculation determines the likelihood that an individual aged 60 survives for 2 years and then dies within the subsequent 2 years. It uses survivor data starting from age 60 to derive the result, yielding a final probability of 0.23. The intermediate step shows a survival probability of 0.88 for the initial deferment phase, confirming the path taken through the life table.

Life table functions dx, qx and px from survivor numbers, `act_life_table_functions`

The numbers dying, the one-year death probabilities and the one-year survival probabilities at each age of a life table given by its survivor numbers lx.

$$d_x = l_x - l_{x+1}, \quad q_x = \frac{d_x}{l_x}, \quad p_x = \frac{l_{x+1}}{l_x}$$

symbol		meaning	SI unit
`lx`	in	survivors at consecutive integer ages (a list)	,
`dx`	out	deaths between ages x and x+1 (a list)	,
`qx`	out	one-year death probabilities (a list)	,
`px`	out	one-year survival probabilities (a list)	,

Applies when: lx non-increasing; only the last lx may be 0.

```
$ actuarial-science solve act_life_table_functions "lx=[1000, 950, 880, 780]"
dx = [50.0,70.0,100.0]
qx = [0.05,0.0736842105263,0.113636363636]
px = [0.95,0.926315789474,0.886363636364]
  by: Life table functions dx, qx and px from survivor numbers  (d_x = l_x - l_(x+1), q_x =
d_x/l_x, p_x = l_(x+1)/l_x)
  assumes: lx non-increasing; only the last lx may be 0
```

What the worked example means: This calculation derives the number of deaths, one-year death probabilities, and one-year survival probabilities from a sequence of survivor counts at consecutive ages. The number of deaths between each age is found by subtracting the survivors at the next age from the current survivors. These death counts are then divided by the initial survivors to determine the probability of dying within that year. Simultaneously, the ratio of survivors at the next age to those at the current age yields the probability of surviving the year. For the provided data, the resulting death counts are [50, 70, 100], the death probabilities are [0.05, 0.0736842, 0.113636], and the survival probabilities are [0.95, 0.926316, 0.886364].

Probability of surviving t years from a life table, `act_survival_probability`

The probability that a life aged x survives t more years, the probability of dying within t years, and the number of deaths between ages x and x+t, from survivor numbers lx starting at age x0.

$$ {}_tp_x = \frac{l_{x+t}}{l_x}, \quad {}_tq_x = \frac{l_x - l_{x+t}}{l_x} $$

symbol		meaning	SI unit
`lx`	in	survivors at consecutive integer ages from x0 (a list)	,
`x0`	in	age of the first entry of lx	,
`x`	in	age of the life	,
`t`	in	number of years	,
`tpx`	out	probability of surviving t years	,
`tqx`	out	probability of dying within t years	,
`deaths`	out	number dying between ages x and x+t	,

Applies when: x0 <= x and x + t inside the table; l_x > 0.

```
$ actuarial-science solve act_survival_probability "lx=[1000, 950, 880, 780]" x0=60 x=61
t=2
tpx = 0.8210526316
tqx = 0.1789473684
deaths = 170
  by: Probability of surviving t years from a life table  (t_p_x = l_(x+t)/l_x; t_q_x = 1 -
t_p_x)
  assumes: x0 <= x and x + t inside the table; l_x > 0
```

What the worked example means: This calculation determines survival and mortality probabilities using a life table. By taking the number of survivors at age 61 plus 2 years and dividing by the number of survivors at age 61, this gives the survival probability 0.821053. The complement of this value gives the probability of death within 2 years, denoted as 0.178947. Additionally, the difference between these survivor counts represents the expected number of deaths, 170, occurring between those ages.

Pure endowment (actuarial discount factor), `act_pure_endowment`

The actuarial present value of 1 paid at time n if a life aged x is then alive, nEx = v^n n_p_x, and the actuarial accumulation factor 1/nEx.

$$ {}_nE_x = v^n \, {}_np_x = v^n \frac{l_{x+n}}{l_x} $$

symbol		meaning	SI unit
`lx`	in	survivors at consecutive integer ages from x0 (a list)	,
`x0`	in	age of the first entry of lx	,
`x`	in	age of the life	,
`n`	in	years until the payment	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`nEx`	out	pure endowment n_E_x = v^n n_p_x	,
`accumulation`	out	actuarial accumulation factor 1/n_E_x	,

Applies when: x + n inside the table; l_(x+n) > 0; 0 < i < 1.

```
$ actuarial-science solve act_pure_endowment "lx=[100000, 99500, 98900, 98200, 97300]"
x0=40 x=40 n=4 i=0.05
nEx = 0.800489508
accumulation = 1.249235612
  by: Pure endowment (actuarial discount factor)  (nEx = v^n l_(x+n)/l_x)
  i = 0.05
  assumes: x + n inside the table; l_(x+n) > 0; 0 < i < 1
```

In the worked example, lx = [100000, 99500, 98900, 98200, 97300], x0 = 40, x = 40, n = 4, i = 0.05 go through the equation above and give nEx = 0.800489508, accumulation = 1.249235612. It applies only when x + n inside the table; l_(x+n) > 0; 0 < i < 1: outside that it says what is wrong instead of returning a number.

Survival to a fractional age (UDD, constant force, Balducci), `act_fractional_age`

The probability of surviving (or dying within) a fraction s of a year from integer age x, and the force of mortality at x+s, under a uniform distribution of deaths, a constant force of mortality or the Balducci assumption.

$$\text{UDD: } {}_sp_x = 1 - sq_x,\ \mu_{x+s} = \frac{q_x}{1 - sq_x};\quad \text{CF: } {}_sp_x = p_x^s;\quad \text{Balducci: } {}_sp_x = \frac{p_x}{1 - (1-s)q_x}$$

symbol		meaning	SI unit
`q`	in	one-year death probability q_x	,
`s`	in	fraction of the year (0 < s < 1)	,
`assumption`	in	fractional-age assumption	,
`spx`	out	probability of surviving to age x+s	,
`sqx`	out	probability of dying before age x+s	,
`mu`	out	force of mortality at age x+s	,

Applies when: 0 < q < 1; 0 < s < 1.

```
$ actuarial-science solve act_fractional_age q=0.1 s=0.5 assumption=udd
spx = 0.95
sqx = 0.05
mu = 0.1052631579
  by: Survival to a fractional age (UDD, constant force, Balducci)  (UDD: s_p_x = 1 - s q,
mu = q/(1 - s q); constant force: (1-q)^s, mu = -ln(1-q); Balducci: (1-q)/(1-(1-s)q), mu =
q/(1-(1-s)q))
  q = 0.1
  s = 0.5
  assumption = udd
  assumes: 0 < q < 1; 0 < s < 1
```

What the worked example means: This calculation determines the probability that a life aged x survives to age x plus fraction s, denoted as 0.95, along with the complementary probability of dying within that interval, 0.05. It also computes the instantaneous force of mortality at age x+s, reported as 0.105263. These values are derived from the one-year death probability 0.1 and the time fraction 0.5 under the selected fractional-age assumption.

Temporary (n-year) expectation of life, `act_temporary_expectation`

The expected number of whole years (curtate) and of years (complete, UDD) lived within the next n years by a life aged x.

$$e_{x:\overline{n}|} = \sum_{k=1}^{n} {}_kp_x, \quad \mathring{e}_{x:\overline{n}|} = \sum_{k=0}^{n-1} \frac{{}_kp_x + {}_{k+1}p_x}{2}$$

symbol		meaning	SI unit
`lx`	in	survivors at consecutive integer ages from x0 (a list)	,
`x0`	in	age of the first entry of lx	,
`x`	in	age of the life	,
`n`	in	term in years	,
`e_curtate`	out	curtate n-year temporary expectation e_(x:n)	,
`e_complete`	out	complete n-year temporary expectation under UDD	,

Applies when: x + n inside the table; n >= 1.

```
$ actuarial-science solve act_temporary_expectation "lx=[1000, 900, 700, 400, 0]" x0=95
x=95 n=2
e_curtate = 1.6
e_complete = 1.75
  by: Temporary (n-year) expectation of life  (e_(x:n) = sum_(k=1..n) k_p_x; e°_(x:n) =
sum_(k=0..n-1) (k_p_x + (k+1)_p_x)/2)
  assumes: x + n inside the table; n >= 1
```

What the worked example means: This calculation determines the expected duration of life within a fixed term. Using survival data starting at age 95, it computes how many full years are lived on average before death or the end of the term, resulting in 1.6. It also estimates the total time lived assuming deaths occur uniformly throughout each year, yielding 1.75. These values represent the partial life expectancy for an individual currently aged 95 over a period of 2 years.

Loss models (16)

Compound Poisson mean, variance and skewness, `act_compound_poisson_moments`

The mean, variance and coefficient of skewness of aggregate losses when the claim count is Poisson(lambda), from the first three raw moments of the claim severity.

$$\kappa_n(S) = \lambda\, E[X^n], \quad \gamma_S = \frac{\lambda E[X^3]}{(\lambda E[X^2])^{3/2}}$$

symbol		meaning	SI unit
`lam`	in	Poisson mean number of claims lambda	,
`m1`	in	E[X]	,
`m2`	in	E[X^2]	,

symbol		meaning	SI unit
`m3`	in	E[X^3]	,
`mean`	out	lambda E[X]	,
`var`	out	lambda E[X^2]	,
`skew`	out	lambda E[X^3] / (lambda E[X^2])^(3/2)	,

Applies when: lambda > 0; moments positive; E[X^2] >= E[X]^2.

```
$ actuarial-science solve act_compound_poisson_moments lam=3 m1=2 m2=6 m3=24
mean = 6
var = 18
skew = 0.9428090416
  by: Compound Poisson mean, variance and skewness  (cumulants of a compound Poisson sum:
kappa_n = lambda E[X^n])
  lam = 3
  m1 = 2
  m2 = 6
  m3 = 24
  assumes: lambda > 0; moments positive; E[X^2] >= E[X]^2
```

What the worked example means: This calculation determines the first three statistical moments of total claims. Using a Poisson claim count with mean 3 and severity moments 2, 6, and 24, this gives the expected total loss as 6. The variability of this total, measured by variance, equals 18. Finally, the asymmetry of the distribution, known as skewness, is calculated as 0.942809, indicating how much the tail extends to the right.

Deductible, policy limit, coinsurance and inflation with exponential losses, `act_exponential_coverage_modifications`

The expected insurer payment per loss and per payment for exponential losses (mean theta) under an ordinary deductible, a policy limit (maximum covered loss), coinsurance and uniform inflation.

$$E[Y^L] = c\,(1+r)\left[E\left[X \wedge \tfrac{u}{1+r}\right] - E\left[X \wedge \tfrac{d}{1+r}\right]\right], \quad E[Y^P] = \frac{E[Y^L]}{S_X\left(\tfrac{d}{1+r}\right)}$$

symbol		meaning	SI unit
`theta`	in	mean loss theta before inflation	,
`d`	in	ordinary deductible	,
`u`	in	policy limit (maximum covered loss)	,
`c`	in	coinsurance fraction paid by the insurer (80% or 0.8)	,
`r`	in	uniform loss inflation (0 for none)	,
`per_loss`	out	expected payment per loss	,
`per_payment`	out	expected payment per payment	,
`max_payment`	out	largest possible payment c (u - d)	,

Applies when: theta > 0; 0 <= d < u; 0 < c <= 1; 0 <= r < 1.

```
$ actuarial-science solve act_exponential_coverage_modifications theta=1000 d=200 u=3000
c=0.8 r=0.1
per_loss = 676.1528531
per_payment = 810.9750963
max_payment = 2240
  by: Deductible, policy limit, coinsurance and inflation with exponential losses  (per
loss = c (1 + r) [E[X ∧ u/(1+r)] - E[X ∧ d/(1+r)]]; per payment = per loss / S(d/(1+r)))
  theta = 1000
  d = 200
  u = 3000
  c = 0.8
  r = 0.1
  assumes: theta > 0; 0 <= d < u; 0 < c <= 1; 0 <= r < 1
```

What the worked example means: This calculation determines the insurer's expected cost for losses that follow an exponential distribution, accounting for an ordinary deductible, a policy limit, coinsurance, and uniform inflation. The expected payment per loss is computed as 676.153, while the expected payment per actual claim made (per payment) is higher at 810.975 because it conditions on losses exceeding the deductible. The largest single amount the insurer could pay in any scenario is capped at 2240, which depends solely on the coinsurance fraction and the gap between the limit and the deductible.

Deductible, policy limit, coinsurance and inflation with Pareto losses, `act_pareto_coverage_modifications`

The expected insurer payment per loss and per payment for two-parameter Pareto losses under an ordinary deductible, a policy limit (maximum covered loss), coinsurance and uniform inflation.

$$E[Y^L] = c(1+r)\left[E\left[X \wedge \tfrac{u}{1+r}\right] - E\left[X \wedge \tfrac{d}{1+r}\right]\right], \quad E[Y^P] = \frac{E[Y^L]}{S_X\left(\tfrac{d}{1+r}\right)}$$

symbol		meaning	SI unit
`alpha`	in	Pareto shape alpha (> 1)	,
`theta`	in	Pareto scale theta before inflation	,
`d`	in	ordinary deductible	,
`u`	in	policy limit (maximum covered loss)	,
`c`	in	coinsurance fraction paid by the insurer (80% or 0.8)	,
`r`	in	uniform loss inflation (0 for none)	,
`per_loss`	out	expected payment per loss	,
`per_payment`	out	expected payment per payment	,
`max_payment`	out	largest possible payment c (u - d)	,

Applies when: alpha > 1; theta > 0; 0 <= d < u; 0 < c <= 1; 0 <= r < 1.

```
$ actuarial-science solve act_pareto_coverage_modifications alpha=3 theta=2000 d=500 u=5000
c=0.8 r=0.05
per_loss = 474.5027461
per_payment = 900.535608
max_payment = 3600
  by: Deductible, policy limit, coinsurance and inflation with Pareto losses  (per loss = c
(1 + r) [E[X ∧ u/(1+r)] - E[X ∧ d/(1+r)]]; per payment = per loss / S(d/(1+r)))
  alpha = 3
  theta = 2000
  d = 500
  u = 5000
  c = 0.8
  r = 0.05
  assumes: alpha > 1; theta > 0; 0 <= d < u; 0 < c <= 1; 0 <= r < 1
```

What the worked example means: This calculation determines the expected cost to the insurer for losses that follow a two-parameter Pareto distribution, accounting for an ordinary deductible, a policy limit, coinsurance, and uniform inflation. The result shows that the average payment per loss event is 474.503, while the average payment conditional on a claim actually being made is 900.536. The highest amount the insurer could ever pay on a single claim is capped at 3600. These figures help actuaries price policies by balancing the frequency of small claims against the severity of large ones under the specified coverage terms.

Individual risk model: aggregate mean and variance, `act_individual_risk_model`

The mean and variance of total claims, and the probability of no claim, for a portfolio of independent policies each paying a fixed benefit b_i with probability q_i.

$$E[S] = \sum_i q_i b_i, \quad \operatorname{Var}(S) = \sum_i q_i (1 - q_i)\, b_i^2$$

symbol		meaning	SI unit
`q`	in	claim probabilities of the policies (a list)	,
`b`	in	benefit amounts of the policies (a list)	,
`mean`	out	expected total claims sum q b	,
`var`	out	variance sum q (1-q) b^2	,
`prob_no_claim`	out	probability of no claim at all	,

Applies when: one benefit per probability; 0 <= q <= 1; b > 0.

```
$ actuarial-science solve act_individual_risk_model "q=[0.01, 0.02, 0.05]" "b=[10000, 5000,
2000]"
mean = 300
var = 1670000
prob_no_claim = 0.92169
  by: Individual risk model: aggregate mean and variance  (E[S] = sum q_i b_i; Var S = sum
q_i (1 - q_i) b_i^2; P(S = 0) = prod (1 - q_i))
  assumes: one benefit per probability; 0 <= q <= 1; b > 0
```

What the worked example means: This calculation determines the aggregate mean and variance for a portfolio of independent policies, where each policy pays a fixed benefit with a specific probability. Using the

provided claim probabilities and benefit amounts, the expected total payout is computed as 300, while the variability of this total is quantified by the variance 1.67e+06. Additionally, the likelihood that none of the policies result in a claim is calculated as 0.92169, providing insight into the frequency of zero-loss years.

Insurer payment for one loss (deductible, limit, coinsurance), `act_single_loss_payment`

The amount the insurer pays and the amount the insured retains for a single loss x under an ordinary deductible d, a policy limit u (maximum covered loss) and coinsurance c.

$$Y = c(\min(X, u) - d)_+$$

symbol		meaning	SI unit
`x`	in	loss amount	,
`d`	in	ordinary deductible	,
`u`	in	policy limit (maximum covered loss)	,
`c`	in	coinsurance fraction paid by the insurer (80% or 0.8)	,
`payment`	out	insurer payment c (min(x, u) - d)+	,
`retained`	out	amount retained by the insured	,

Applies when: x, d >= 0; u > d; 0 < c <= 1.

```
$ actuarial-science solve act_single_loss_payment x=5000 d=500 u=4000 c=0.8
payment = 2800
retained = 2200
  by: Insurer payment for one loss (deductible, limit, coinsurance)  (payment = c x (min(x,
u) - d)+)
  x = 5000
  d = 500
  u = 4000
  c = 0.8
  assumes: x, d >= 0; u > d; 0 < c <= 1
```

What the worked example means: This calculation determines the financial split between the insurer and the policyholder for a single claim. Starting with a reported loss amount of 5000, the ordinary deductible of 500 is subtracted, but only up to the policy limit of 4000. The resulting covered amount is then multiplied by the coinsurance fraction of 0.8 to find the insurer's payment, which is 2800. The remainder, representing what the insured must pay out of pocket, is 2200. This ensures that neither party pays more than their contractual share based on the specific terms of the policy.

Mean and variance of aggregate losses (compound distribution), `act_compound_distribution_moments`

The mean and variance of aggregate losses S = X_1 + ... + X_N from the mean and variance of the claim count N and of the claim severity X (independent, identically distributed severities).

$$E[S] = E[N]\,E[X], \quad \operatorname{Var}(S) = E[N]\operatorname{Var}(X) + \operatorname{Var}(N)\,E[X]^2$$

symbol		meaning	SI unit
`EN`	in	mean number of claims E[N]	,
`VarN`	in	variance of the number of claims	,
`EX`	in	mean claim severity E[X]	,
`VarX`	in	variance of the claim severity	,
`ES`	out	mean aggregate loss E[N] E[X]	,
`VarS`	out	variance E[N] Var X + Var N E[X]^2	,

Applies when: moments non-negative; N and the X's independent.

```
$ actuarial-science solve act_compound_distribution_moments EN=3 VarN=3 EX=1000 VarX=250000
ES = 3000
VarS = 3750000
  by: Mean and variance of aggregate losses (compound distribution)  (E[S] = E[N] E[X]; Var
S = E[N] Var X + Var N (E[X])^2)
  EN = 3
  VarN = 3
  EX = 1000
  VarX = 250000
  assumes: moments non-negative; N and the X's independent
```

What the worked example means: This calculation determines the expected total cost and variability of claims. The mean aggregate loss, shown as 3000, equals the average number of claims multiplied by the average severity per claim. The variance of aggregate losses, shown as 3.75e+06, combines the variability from both the frequency of claims and the size of individual claims. This result helps insurers understand the overall financial exposure by accounting for how many claims occur and how large each claim tends to be.

Negative binomial claim frequency probabilities, `act_negative_binomial_frequency`

The probability of exactly k claims and of at most k claims, with the mean and variance, for a negative binomial claim count with parameters r and beta (mean r beta).

$$P(N = k) = \binom{r + k - 1}{k} \left(\frac{1}{1+\beta}\right)^{r} \left(\frac{\beta}{1+\beta}\right)^{k}$$

symbol		meaning	SI unit
`r`	in	negative binomial r	,
`beta`	in	negative binomial beta	,
`k`	in	number of claims	,
`pmf`	out	P(N = k)	,
`cdf`	out	P(N <= k)	,
`mean`	out	mean r beta	,
`var`	out	variance r beta (1 + beta)	,

Applies when: r, beta > 0; k >= 0.

```
$ actuarial-science solve act_negative_binomial_frequency r=2 beta=1.5 k=3
pmf = 0.13824
cdf = 0.66304
mean = 3
var = 7.5
  by: Negative binomial claim frequency probabilities  (P(N = k) = Gamma(r+k)/(Gamma(r) k!) (1/(1+beta))^r (beta/(1+beta))^k)
  r = 2
  beta = 1.5
  assumes: r, beta > 0; k >= 0
```

What the worked example means: This calculation determines the likelihood of observing a specific number of insurance claims using the negative binomial distribution. Given the shape parameter 2 and scale parameter 1.5, the model computes the probability mass for exactly 3 claims, resulting in 0.13824. It also provides the cumulative probability that the claim count does not exceed 3, which is 0.66304. Additionally, the expected mean number of claims is calculated as 3, and the variance describing the spread of claim counts is 7.5. These metrics help actuaries understand both the central tendency and the dispersion of risk.

Normal approximation to aggregate losses, `act_normal_approximation_aggregate`

The probability that aggregate losses exceed an amount s under the normal approximation with the given mean and variance.

$$P(S > s) \approx 1 - \Phi\left(\frac{s - E[S]}{\sqrt{\mathrm{Var}(S)}}\right)$$

symbol		meaning	SI unit
`mean`	in	mean of aggregate losses	,
`var`	in	variance of aggregate losses	,
`s`	in	threshold amount	,
`prob_exceed`	out	P(S > s) by the normal approximation	,
`z`	out	standardized threshold (s - mean)/sd	,

Applies when: variance > 0.

```
$ actuarial-science solve act_normal_approximation_aggregate mean=3000 var=1000000 s=5000
prob_exceed = 0.02275013195
z = 2
  by: Normal approximation to aggregate losses  (P(S > s) = 1 - Phi((s - E[S])/sd(S)))
  mean = 3000
  var = 1000000
  s = 5000
  assumes: variance > 0
```

What the worked example means: This calculation determines the likelihood that total losses surpass a specific threshold by applying the normal approximation. It uses the provided mean aggregate loss of 3000 and variance of 1e+06 to standardize the target amount 5000, resulting in a z-score of 2. The final output, 0.0227501, represents the tail probability under this distribution, indicating how extreme the threshold is relative to the expected loss variability.

Ordinary deductible with exponential losses, `act_exponential_deductible`

The expected payment per loss and per payment, the limited expected value and the loss elimination ratio of an ordinary deductible d when losses are exponential with mean theta.

$$E[X \wedge d] = \theta\left(1 - e^{-d/\theta}\right), \quad E[(X-d)_+] = \theta e^{-d/\theta}, \quad e(d) = \theta$$

symbol		meaning	SI unit
`theta`	in	mean loss theta	,
`d`	in	ordinary deductible	,
`mean`	out	mean loss E[X]	,
`lev_d`	out	limited expected value E[X ∧ d]	,
`per_loss`	out	expected payment per loss E[(X - d)+]	,
`per_payment`	out	expected payment per payment (mean excess loss e(d))	,
`LER`	out	loss elimination ratio E[X ∧ d]/E[X]	,

Applies when: theta > 0; d >= 0.

```
$ actuarial-science solve act_exponential_deductible theta=1000 d=500
mean = 1000
lev_d = 393.4693403
per_loss = 606.5306597
per_payment = 1000
LER = 0.3934693403
  by: Ordinary deductible with exponential losses  (per loss = E[X] - E[X ∧ d]; per payment
= per loss / S(d); LER = E[X ∧ d]/E[X]; E[X ∧ d] = theta (1 - e^(-d/theta)))
  theta = 1000
  d = 500
  assumes: theta > 0; d >= 0
```

What the worked example means: This calculation determines key insurance metrics for an ordinary deductible applied to exponential losses. Given a mean loss of 1000 and a deductible of 500, it computes the limited expected value as 393.469, which represents the average portion of each loss absorbed by the insured. The expected payment per loss is found to be 606.531, while the expected payment per actual claim is 1000. Additionally, the loss elimination ratio is calculated as 0.393469, indicating the fraction of total expected loss eliminated by the deductible.

Ordinary deductible with lognormal losses, `act_lognormal_deductible`

The expected payment per loss and per payment, the limited expected value and the loss elimination ratio of an ordinary deductible d when losses are lognormal (mu, sigma of the log of the loss).

$$E[X \wedge d] = e^{\mu+\sigma^2/2}\,\Phi\left(\frac{\ln d - \mu - \sigma^2}{\sigma}\right) + d\left[1 - \Phi\left(\frac{\ln d - \mu}{\sigma}\right)\right]$$

symbol		meaning	SI unit
`mu`	in	mean of the log of the loss	,

symbol		meaning	SI unit
`sigma`	in	standard deviation of the log of the loss	,
`d`	in	ordinary deductible	,
`mean`	out	mean loss E[X]	,
`lev_d`	out	limited expected value E[X ∧ d]	,
`per_loss`	out	expected payment per loss E[(X - d)+]	,
`per_payment`	out	expected payment per payment (mean excess loss e(d))	,
`LER`	out	loss elimination ratio E[X ∧ d]/E[X]	,

Applies when: sigma > 0; d >= 0.

```
$ actuarial-science solve act_lognormal_deductible mu=7 sigma=1 d=1000
mean = 1808.042414
lev_d = 785.1059202
per_loss = 1022.936494
per_payment = 1905.803319
LER = 0.4342298134
  by: Ordinary deductible with lognormal losses  (per loss = E[X] - E[X ∧ d]; per payment =
per loss / S(d); LER = E[X ∧ d]/E[X]; E[X ∧ d] = e^(mu + sigma^2/2) Phi((ln d - mu -
sigma^2)/sigma) + d [1 - Phi((ln d - mu)/sigma)])
  mu = 7
  sigma = 1
  d = 1000
  assumes: sigma > 0; d >= 0
```

What the worked example means: This calculation determines insurance metrics for losses following a lognormal distribution with log-mean 7 and log-standard-deviation 1, subject to an ordinary deductible of 1000. The mean loss is computed as 1808.04. The limited expected value, representing the average amount of loss covered up to the deductible, is 785.106. Consequently, the expected payment per loss event is 1022.94. When a payment actually occurs, the expected payment per payment is 1905.8. Finally, the loss elimination ratio, indicating the proportion of total expected loss removed by the deductible, is 0.43423.

Ordinary deductible with Pareto losses, `act_pareto_deductible`

The expected payment per loss and per payment, the limited expected value and the loss elimination ratio of an ordinary deductible d when losses follow a two-parameter Pareto distribution (alpha, theta).

$$E[X \wedge d] = \frac{\theta}{\alpha - 1}\left[1 - \left(\frac{\theta}{\theta + d}\right)^{\alpha - 1}\right], \quad e(d) = \frac{\theta + d}{\alpha - 1}$$

symbol		meaning	SI unit
`alpha`	in	Pareto shape alpha (> 1)	,
`theta`	in	Pareto scale theta	,
`d`	in	ordinary deductible	,
`mean`	out	mean loss E[X]	,
`lev_d`	out	limited expected value E[X ∧ d]	,

symbol		meaning	SI unit
`per_loss`	out	expected payment per loss E[(X - d)+]	,
`per_payment`	out	expected payment per payment (mean excess loss e(d))	,
`LER`	out	loss elimination ratio E[X ∧ d]/E[X]	,

Applies when: alpha > 1; theta > 0; d >= 0.

```
$ actuarial-science solve act_pareto_deductible alpha=3 theta=2000 d=500
mean = 1000
lev_d = 360
per_loss = 640
per_payment = 1250
LER = 0.36
  by: Ordinary deductible with Pareto losses  (per loss = E[X] - E[X ∧ d]; per payment =
per loss / S(d); LER = E[X ∧ d]/E[X]; E[X ∧ d] = theta/(alpha-1) [1 -
(theta/(theta+d))^(alpha-1)])
  alpha = 3
  theta = 2000
  d = 500
  assumes: alpha > 1; theta > 0; d >= 0
```

What the worked example means: This calculation determines insurance payout metrics for losses following a Pareto distribution with shape 3 and scale 2000, subject to an ordinary deductible of 500. It computes the limited expected value at 360 and the overall mean loss at 1000. From these, the expected payment per loss is derived as 640, while the expected payment per payment is 1250. The loss elimination ratio, representing the proportion of total loss removed by the deductible, is calculated as 0.36. These figures help insurers estimate costs and adjust premiums based on the specific risk profile defined by the input parameters.

Panjer recursion for a compound negative binomial distribution, `act_panjer_negative_binomial`

The probabilities of aggregate losses 0, 1, ..., smax units for a negative binomial (r, beta) number of claims with claim severities on 0, 1, 2, ... units, by Panjer's recursion, and P(S <= smax).

$$g_s = \frac{1}{1 - a f_0} \sum_{j=1}^{s} \left(a + \frac{bj}{s} \right) f_j \, g_{s-j}, \quad a = \frac{\beta}{1+\beta}, \; b = (r-1)a$$

symbol		meaning	SI unit
`r`	in	negative binomial r	,
`beta`	in	negative binomial beta	,
`f`	in	severity probabilities of 0, 1, 2, ... units (a list adding to 1)	,
`smax`	in	largest aggregate amount wanted	,
`g`	out	P(S = s) for s = 0..smax (a list)	,
`F_smax`	out	P(S <= smax)	,

Applies when: r, beta > 0; severity probabilities add to 1; smax <= 300.

```
$ actuarial-science solve act_panjer_negative_binomial r=2 beta=1 "f=[0, 0.5, 0.3, 0.2]"
smax=5
g = [0.25,0.125,0.121875,0.121875,0.0873828125,0.07130859375]
F_smax = 0.7774414063
  by: Panjer recursion for a compound negative binomial distribution  ((a, b, 0) class: a =
beta/(1+beta), b = (r-1) a; g_0 = (1 + beta(1 - f_0))^(-r); g_s = [sum (a + b j/s) f_j
g_(s-j)]/(1 - a f_0))
  r = 2
  beta = 1
  assumes: r, beta > 0; severity probabilities add to 1; smax <= 300
```

What the worked example means: This calculation determines the probabilities of total losses for a compound negative binomial model using Panjer recursion. Given the shape parameter r, scale parameter beta, and severity probabilities f, it computes the distribution of aggregate claims up to smax. The resulting cumulative probability that total losses do not exceed smax is 0.777441. This value reflects the combined impact of claim frequency variability controlled by r and beta, and the specific severity profile defined in f.

Panjer recursion for a compound Poisson distribution, `act_panjer_poisson`

The probabilities of aggregate losses 0, 1, ..., smax (in monetary units) for a Poisson(lambda) number of claims with claim severities on 0, 1, 2, ... units, by Panjer's recursion, and P(S <= smax).

$$g_0 = e^{-\lambda(1-f_0)}, \quad g_s = \frac{\lambda}{s}\sum_{j=1}^{s} j\, f_j\, g_{s-j}$$

symbol		meaning	SI unit
`lam`	in	Poisson mean number of claims lambda	,
`f`	in	severity probabilities of 0, 1, 2, ... units (a list adding to 1)	,
`smax`	in	largest aggregate amount wanted	,
`g`	out	P(S = s) for s = 0..smax (a list)	,
`F_smax`	out	P(S <= smax)	,

Applies when: 0 < lambda <= 200; severity probabilities add to 1; smax <= 300.

```
$ actuarial-science solve act_panjer_poisson lam=2 "f=[0, 0.5, 0.3, 0.2]" smax=5
g =
[0.135335283237,0.135335283237,0.14886881156,0.157891163776,0.124734019383,0.0985691979573]
F_smax = 0.8007337592
  by: Panjer recursion for a compound Poisson distribution  (g_0 = e^(-lambda(1 - f_0));
g_s = (1/s) sum_(j=1..s) lambda j f_j g_(s-j))
  lam = 2
  assumes: 0 < lambda <= 200; severity probabilities add to 1; smax <= 300
```

What the worked example means: This calculation determines the distribution of total losses by applying Panjer's recursion to a compound Poisson process. Given a mean claim count of 2 and severity probabilities defined over discrete units, the method iteratively computes the probability mass function for each possible aggregate amount up to 5. The final output includes the full list of individual probabilities and the cumulative chance that total losses do not exceed this maximum threshold, which is reported as 0.800734.

Poisson claim frequency probabilities, `act_poisson_frequency`

The probability of exactly k claims and of at most k claims when the number of claims is Poisson with mean lambda.

$$P(N = k) = \frac{e^{-\lambda}\lambda^{k}}{k!}$$

symbol		meaning	SI unit
`lam`	in	Poisson mean number of claims lambda	,
`k`	in	number of claims	,
`pmf`	out	P(N = k)	,
`cdf`	out	P(N <= k)	,
`mean`	out	mean number of claims	,
`var`	out	variance of the number of claims	,

Applies when: 0 < lambda <= 500; k >= 0.

```
$ actuarial-science solve act_poisson_frequency lam=2 k=3
pmf = 0.1804470443
cdf = 0.8571234605
mean = 2
var = 2
  by: Poisson claim frequency probabilities  (P(N = k) = e^(-lambda) lambda^k / k!)
  lam = 2
  assumes: 0 < lambda <= 500; k >= 0
```

What the worked example means: This calculation determines the likelihood of observing a specific number of insurance claims using the Poisson distribution. Given an expected claim frequency of 2, this gives the exact probability that the actual count equals 3, which results in 0.180447. Additionally, it provides the cumulative probability that the claim count does not exceed 3, yielding 0.857123. The model also confirms that the mean number of claims is 2 and the variance is 2, reflecting the defining property of the Poisson distribution where these two statistics are equal.

Premium principles: expected value, variance, standard deviation, `act_premium_principles`

The premium for a risk with a given mean and variance of loss by the expected value principle (1 + theta) E[X], the variance principle E[X] + theta Var X or the standard deviation principle E[X] + theta sd(X).

$$\pi = (1+\theta)E[X], \quad \pi = E[X] + \theta\,\mathrm{Var}(X), \quad \pi = E[X] + \theta\sqrt{\mathrm{Var}(X)}$$

symbol		meaning	SI unit
`mean`	in	expected loss	,
`var`	in	variance of the loss	,
`theta`	in	loading factor	,
`principle`	in	premium principle	,

symbol		meaning	SI unit
`premium`	out	premium	,
`loading`	out	risk loading added to the expected loss	,

Applies when: mean, variance, loading >= 0.

```
$ actuarial-science solve act_premium_principles mean=1000 var=250000 theta=0.2
principle=expected_value
premium = 1200
loading = 200
  by: Premium principles: expected value, variance, standard deviation  (expected value: (1
+ theta) E[X]; variance: E[X] + theta Var X; standard deviation: E[X] + theta sd(X))
  mean = 1000
  var = 250000
  theta = 0.2
  principle = expected_value
  assumes: mean, variance, loading >= 0
```

What the worked example means: This calculation determines the insurance premium based on the expected loss, variance, and a loading factor. Depending on the chosen principle, the result reflects either the mean multiplied by one plus theta, or the mean added to theta times the variance or standard deviation. The final premium is 1200, which includes a risk loading component of 200 above the pure expected loss.

Stop-loss premium for a compound Poisson distribution, `act_stop_loss_premium`

The net stop-loss premium E[(S - d)+] for a retention of d units, with E[S] and the limited expectation E[S ∧ d], when S is compound Poisson with claim severities on 0, 1, 2, ... units.

$$E[(S-d)_+] = E[S] - \sum_{s=0}^{d-1} (1 - F_S(s))$$

symbol		meaning	SI unit
`lam`	in	Poisson mean number of claims lambda	,
`f`	in	severity probabilities of 0, 1, 2, ... units (a list adding to 1)	,
`d`	in	retention (whole number of units)	,
`premium`	out	net stop-loss premium E[(S - d)+]	,
`ES`	out	expected aggregate loss E[S]	,
`limited`	out	limited expectation E[S ∧ d]	,

Applies when: 0 < lambda <= 200; severity probabilities add to 1; whole-number retention.

```
$ actuarial-science solve act_stop_loss_premium lam=2 "f=[0, 0.5, 0.3, 0.2]" d=3
premium = 1.225545228
ES = 3.4
limited = 2.174454772
  by: Stop-loss premium for a compound Poisson distribution  (E[(S - d)+] = E[S] -
sum_(s=0..d-1) P(S > s), with P(S = s) from Panjer's recursion)
  lam = 2
  assumes: 0 < lambda <= 200; severity probabilities add to 1; whole-number retention
```

What the worked example means: This calculation determines the net stop-loss premium for a retention level of 3 units, assuming the aggregate loss follows a compound Poisson distribution. The result, 1.22555, represents the expected amount paid by the insurer above the retention. It is derived from the total expected loss 3.4 minus the limited expectation 2.17445, which captures the average loss capped at the retention point. The severity probabilities provided in [0, 0.5, 0.3, 0.2] and the claim frequency parameter 2 define the underlying distribution used to compute these values.

Loss reserving (7)

Bornhuetter-Ferguson reserve, `act_bornhuetter_ferguson`

The Bornhuetter-Ferguson reserve and ultimate losses of each origin year from a cumulative loss triangle, the earned premiums and an expected loss ratio: the unreported part (1 - 1/CDF) of the expected losses, with chain-ladder volume-weighted factors.

$$R_i^{BF} = P_i \cdot \mathrm{ELR}\left(1 - \frac{1}{F_{N-i}}\right), \qquad \hat{U}_i = C_{i,N-i} + R_i^{BF}$$

symbol		meaning	SI unit
`T`	in	cumulative loss triangle: origin years in rows, development ages in columns, unknown cells written as 0	,
`P`	in	earned premium of each origin year (a list)	,
`elr`	in	expected (a priori) loss ratio (70% or 0.7)	,
`ultimate`	out	ultimate losses of each origin year (a list)	,
`reserve`	out	reserve of each origin year (a list)	,
`total_reserve`	out	total reserve	,

Applies when: square triangle; one positive premium per origin year; 0 < ELR <= 500%.

```
$ actuarial-science solve act_bornhuetter_ferguson "T=[[100, 150, 175], [110, 168, 0],
[120, 0, 0]]" "P=[200, 210, 220]" elr=0.8
ultimate = [175.0,192.0,196.377358491]
reserve = [0.0,24.0,76.3773584906]
total_reserve = 100.3773585
  by: Bornhuetter-Ferguson reserve  (reserve_i = P_i ELR (1 - 1/CDF_i); ultimate_i =
latest_i + reserve_i)
  elr = 0.8
  assumes: square triangle; one positive premium per origin year; 0 < ELR <= 500%
```

What the worked example means: This calculation determines the unpaid claims reserve and projected ultimate losses for a specific accident year using the Bornhuetter-Ferguson approach. It combines the actual reported losses to date with an estimate of unreported losses derived from the earned premium and an assumed expected loss ratio. The reserve is calculated as the product of the premium [200, 210, 220], the expected loss ratio 0.8, and the proportion of losses not yet developed. This result is then added to the latest cumulative paid or incurred losses to form the total ultimate loss estimate of [175, 192, 196.377]. The resulting reserve figure of [0, 24, 76.3774] represents the liability for claims that have occurred but are not yet fully settled.

Cape Cod (Stanard-Buhlmann) reserve, `act_cape_cod`

The Cape Cod expected loss ratio (reported losses over used-up premium), and the resulting reserves and ultimate losses of each origin year, from a cumulative loss triangle and the earned premiums.

$$\mathrm{ELR}_{CC} = \frac{\sum_i C_{i,N-i}}{\sum_i P_i / F_{N-i}}, \qquad R_i = P_i \, \mathrm{ELR}_{CC} \left(1 - \frac{1}{F_{N-i}}\right)$$

symbol		meaning	SI unit
`T`	in	cumulative loss triangle: origin years in rows, development ages in columns, unknown cells written as 0	,
`P`	in	earned premium of each origin year (a list)	,
`elr`	out	Cape Cod expected loss ratio	,
`ultimate`	out	ultimate losses of each origin year (a list)	,
`reserve`	out	reserve of each origin year (a list)	,
`total_reserve`	out	total reserve	,

Applies when: square triangle; one positive premium per origin year; no trend, no decay.

```
$ actuarial-science solve act_cape_cod "T=[[100, 150, 175], [110, 168, 0], [120, 0, 0]]"
"P=[200, 210, 220]"
elr = 0.9176888556
ultimate = [175.0,195.530665669,207.613313388]
reserve = [0.0,27.5306656694,87.6133133882]
total_reserve = 115.1439791
  by: Cape Cod (Stanard-Buhlmann) reserve  (ELR = sum latest_i / sum (P_i / CDF_i);
reserve_i = P_i ELR (1 - 1/CDF_i))
  assumes: square triangle; one positive premium per origin year; no trend, no decay
```

What the worked example means: This calculation determines the Cape Cod expected loss ratio by dividing the sum of latest reported losses by the sum of used-up premiums, resulting in an estimate of 0.917689. Using this ratio, ultimate losses for each origin year are projected as [175, 195.531, 207.613], and the unpaid claim reserves are calculated as [0, 27.5307, 87.6133], which sum to a total reserve of 115.144. This method assumes that the expected loss ratio is constant across all years after adjusting for exposure.

Chain-ladder age-to-age and cumulative development factors, `act_chain_ladder_factors`

The volume-weighted age-to-age development factors of a cumulative loss triangle and the cumulative factors to ultimate from each development age (no tail factor).

$$\hat{f}_j = \frac{\sum_i C_{i,j+1}}{\sum_i C_{i,j}}, \qquad F_j = \prod_{k \geq j} \hat{f}_k$$

symbol		meaning	SI unit
`T`	in	cumulative loss triangle: origin years in rows, development ages in columns, unknown cells written as 0	,
`ldf`	out	age-to-age factors f_1 ... f_(N-1) (a list)	,
`cdf`	out	cumulative development factors to ultimate from each age, last = 1 (a list)	,

Applies when: square triangle with N >= 2; known cells positive, unknown cells 0; no tail factor.

```
$ actuarial-science solve act_chain_ladder_factors "T=[[100, 150, 175], [110, 168, 0],
[120, 0, 0]]"
ldf = [1.51428571429,1.16666666667]
cdf = [1.76666666667,1.16666666667,1.0]
  by: Chain-ladder age-to-age and cumulative development factors  (f_j = sum_i C_(i,j+1) /
sum_i C_(i,j) over origins with both ages known; CDF_j = product of f from age j on)
  assumes: square triangle with N >= 2; known cells positive, unknown cells 0; no tail
factor
```

What the worked example means: This calculation derives volume-weighted age-to-age development factors from a cumulative loss triangle, where each factor represents the ratio of total losses at a later development age to total losses at the preceding age. The resulting list of factors, [1.51429, 1.16667], shows how claims develop year by year. These individual factors are then multiplied together to produce the cumulative development factors, [1.76667, 1.16667, 1], which indicate the expected multiple by which current reported losses must grow to reach ultimate losses from each specific development age.

Chain-ladder ultimate losses and reserves, `act_chain_ladder_reserve`

The ultimate losses, the reserve (ultimate less reported to date) of each origin year and the total reserve of a cumulative loss triangle by the volume-weighted chain-ladder method (no tail factor).

$$\hat{C}_{i,N} = C_{i,N-i} \prod_{k \geq N-i} \hat{f}_k, \qquad R_i = \hat{C}_{i,N} - C_{i,N-i}$$

symbol		meaning	SI unit
`T`	in	cumulative loss triangle: origin years in rows, development ages in columns, unknown cells written as 0	,
`ultimate`	out	ultimate losses of each origin year (a list)	,
`reserve`	out	reserve of each origin year (a list)	,
`total_reserve`	out	total reserve	,

Applies when: square triangle with N >= 2; known cells positive, unknown cells 0; no tail factor.

```
$ actuarial-science solve act_chain_ladder_reserve "T=[[100, 150, 175], [110, 168, 0],
[120, 0, 0]]"
ultimate = [175.0,196.0,212.0]
reserve = [0.0,28.0,92.0]
total_reserve = 120
  by: Chain-ladder ultimate losses and reserves  (ultimate_i = latest_i x CDF at its age;
reserve_i = ultimate_i - latest_i)
  assumes: square triangle with N >= 2; known cells positive, unknown cells 0; no tail
factor
```

What the worked example means: This calculation estimates the final cost of claims for each accident year by projecting current cumulative losses to their ultimate value using volume-weighted development factors. The result, [175, 196, 212], represents the expected total paid and incurred losses for each cohort. By subtracting the latest reported figures from these projections, this gives the reserve, [0, 28, 92], which is the amount set aside to cover future claim payments. The sum of these individual reserves yields the total reserve liability, 120, ensuring sufficient funds are held against outstanding obligations.

Chain-ladder with simple-average development factors, `act_chain_ladder_simple_average`

The simple-average age-to-age factors (mean of the individual link ratios), the ultimate losses of each origin year and the total reserve of a cumulative loss triangle (no tail factor).

$$\hat{f}_j = \frac{1}{n_j} \sum_i \frac{C_{i,j+1}}{C_{i,j}}$$

symbol		meaning	SI unit
`T`	in	cumulative loss triangle: origin years in rows, development ages in columns, unknown cells written as 0	,
`ldf`	out	simple-average age-to-age factors (a list)	,
`ultimate`	out	ultimate losses of each origin year (a list)	,
`total_reserve`	out	total reserve	,

Applies when: square triangle with N >= 2; known cells positive, unknown cells 0; no tail factor.

```
$ actuarial-science solve act_chain_ladder_simple_average "T=[[100, 150, 175], [110, 168,
0], [120, 0, 0]]"
ldf = [1.51363636364,1.16666666667]
ultimate = [175.0,196.0,211.909090909]
total_reserve = 119.9090909
  by: Chain-ladder with simple-average development factors  (f_j = mean over origins of
C_(i,j+1)/C_(i,j); ultimate_i = latest_i x product of f)
  assumes: square triangle with N >= 2; known cells positive, unknown cells 0; no tail
factor
```

What the worked example means: This calculation derives simple-average age-to-age factors from the cumulative loss triangle [100 150 175; 110 168 0; 120 0 0], then projects each origin year to its ultimate value. The resulting ultimate losses are [175, 196, 211.909], which yield a total reserve of 119.909. This reserve

represents the expected future payments needed to settle all claims, assuming no tail factor beyond the latest development age.

Expected claims reserving method, `act_expected_claims_method`

The ultimate losses (premium x expected loss ratio), the reserve (ultimate less paid or reported to date) of each origin year and the total reserve by the expected claims method.

$$\hat{U}_i = P_i \cdot \mathrm{ELR}, \qquad R_i = \hat{U}_i - C_i$$

symbol		meaning	SI unit
`P`	in	earned premium of each origin year (a list)	,
`elr`	in	expected loss ratio (65% or 0.65)	,
`paid`	in	losses paid (or reported) to date of each origin year (a list)	,
`ultimate`	out	expected ultimate losses (a list)	,
`reserve`	out	reserve of each origin year (a list)	,
`total_reserve`	out	total reserve	,

Applies when: one paid amount per premium; 0 < ELR <= 500%.

```
$ actuarial-science solve act_expected_claims_method "P=[1000, 1100, 1200]" elr=0.7 "paid=
[650, 500, 300]"
ultimate = [700.0,770.0,840.0]
reserve = [50.0,270.0,540.0]
total_reserve = 860
  by: Expected claims reserving method  (ultimate_i = P_i x ELR; reserve_i = ultimate_i -
paid_i)
  elr = 0.7
  assumes: one paid amount per premium; 0 < ELR <= 500%
```

What the worked example means: This calculation determines the expected ultimate losses for each origin year by multiplying the earned premium by the expected loss ratio. The reserve for each year is then found by subtracting the losses already paid from these expected ultimate figures. Summing these individual reserves yields the total reserve, which stands at 860. This represents the estimated liability for future payments based on current experience.

Mack's standard error of chain-ladder reserves, `act_mack_standard_error`

The standard errors (root mean squared errors of prediction) of the chain-ladder reserve of each origin year and of the total reserve by Mack's distribution-free model, with the variance parameters sigma^2 of each development age.

$$\widehat{\mathrm{mse}}(\hat{R}_i) = \hat{C}_{iN}^2 \sum_{k=N-i}^{N-1} \frac{\hat{\sigma}_k^2}{\hat{f}_k^2} \left(\frac{1}{\hat{C}_{ik}} + \frac{1}{\sum_j C_{jk}} \right)$$

symbol		meaning	SI unit
`T`	in	cumulative loss triangle: origin years in rows, development ages in columns, unknown cells written as 0	,
`se`	out	standard error of each origin year's reserve (a list)	,
`total_se`	out	standard error of the total reserve	,
`sigma2`	out	Mack variance parameters sigma^2 of each age (a list)	,

Applies when: square triangle with N >= 4; known cells positive, unknown cells 0; last sigma^2 by Mack's extrapolation.

```
$ actuarial-science solve act_mack_standard_error "T=[[1000, 1800, 2100, 2200], [1100,
2000, 2300, 0], [1200, 2100, 0, 0], [1300, 0, 0, 0]]"
se = [0.0,15.2267485994,34.5563923039,72.2173151114]
total_se = 90.68083258
sigma2 = [1.43939393939,0.263157894737,0.0481119696749]
  by: Mack's standard error of chain-ladder reserves  (sigma_k^2 = sum C_ik (C_(i,k+1)/C_ik
- f_k)^2/(n_k - 1); mse(R_i) = C_iN^2 sum_k sigma_k^2/f_k^2 (1/C_ik + 1/S_k); total adds
the covariance terms)
  assumes: square triangle with N >= 4; known cells positive, unknown cells 0; last sigma^2
by Mack's extrapolation
```

What the worked example means: This calculation determines the uncertainty of chain-ladder reserve estimates using Mack's distribution-free model. It computes variance parameters for each development age, denoted as [1.43939, 0.263158, 0.048112], based on the residuals in the cumulative loss triangle [1000 1800 2100 2200; 1100 2000 2300 0; 1200 2100 0 0; 1300 0 0 0]. These parameters are then used to derive the standard error for each origin year and the aggregate total reserve, reported here as 90.6808. The result quantifies the prediction error expected if the historical development pattern were repeated under similar conditions.

Multiple decrements (2)

Associated single decrement rates from a multiple decrement table (UDD), `act_multiple_decrement_to_single`

The associated single decrement probabilities q'(1) and q'(2) from the two decrement probabilities of a double decrement table, assuming each decrement is uniformly distributed within the year in the multiple decrement table.

$$q_x'^{(j)} = 1 - \left(p_x^{(\tau)}\right)^{q_x^{(j)}/q_x^{(\tau)}}$$

symbol		meaning	SI unit
`q1`	in	probability of leaving by decrement 1 within the year q(1)	,
`q2`	in	probability of leaving by decrement 2 within the year q(2)	,
`q1_single`	out	associated single decrement probability q'(1)	,
`q2_single`	out	associated single decrement probability q'(2)	,
`q_total`	out	total decrement probability q(tau)	,

Applies when: q1, q2 > 0; q1 + q2 < 1; UDD in the multiple decrement table.

```
$ actuarial-science solve act_multiple_decrement_to_single q1=0.1 q2=0.2
q1_single = 0.1120959983
q2_single = 0.2116264837
q_total = 0.3
  by: Associated single decrement rates from a multiple decrement table (UDD)  (q'(j) = 1 -
(p(tau))^(q(j)/q(tau)))
  q1 = 0.1
  q2 = 0.2
  assumes: q1, q2 > 0; q1 + q2 < 1; UDD in the multiple decrement table
```

What the worked example means: This calculation determines the independent rates of decrement by removing the interaction between competing risks. Given a multiple decrement table with probabilities 0.1 for the first cause and 0.2 for the second, decrements are assumed to be uniformly distributed within the year. The resulting associated single decrement probability for the first cause is 0.112096, while the total probability of leaving due to any cause is 0.3. This method isolates the effect of each specific cause as if it were the only one operating.

Multiple decrement probabilities from single decrement rates (UDD), `act_single_to_multiple_decrement`

The double decrement probabilities q(1) and q(2) from the associated single decrement probabilities q'(1) and q'(2), each uniformly distributed within the year in its single decrement table.

$$q_x^{(1)} = q_x'^{(1)}\left(1 - \tfrac{1}{2}q_x'^{(2)}\right), \quad q_x^{(2)} = q_x'^{(2)}\left(1 - \tfrac{1}{2}q_x'^{(1)}\right)$$

symbol		meaning	SI unit
`q1s`	in	associated single decrement probability q'(1)	,
`q2s`	in	associated single decrement probability q'(2)	,
`q1`	out	probability of leaving by decrement 1, q(1)	,
`q2`	out	probability of leaving by decrement 2, q(2)	,
`q_total`	out	total decrement probability q(tau)	,

Applies when: 0 < q'(1), q'(2) < 1; UDD in each single decrement table.

```
$ actuarial-science solve act_single_to_multiple_decrement q1s=0.1 q2s=0.2
q1 = 0.09
q2 = 0.19
q_total = 0.28
  by: Multiple decrement probabilities from single decrement rates (UDD)  (q(1) = q'(1) (1
- q'(2)/2), q(2) = q'(2) (1 - q'(1)/2))
  q1s = 0.1
  q2s = 0.2
  assumes: 0 < q'(1), q'(2) < 1; UDD in each single decrement table
```

What the worked example means: This calculation converts associated single decrement probabilities into multiple decrement probabilities under the assumption that each decrement is uniformly distributed within the year. Given an associated single decrement probability for cause one of 0.1 and for cause two of 0.2, the

resulting dependent probability for cause one is 0.09 and for cause two is 0.19. The sum of these individual probabilities gives the total probability of exiting the population during the period, which is 0.28. This method accounts for the fact that individuals can leave due to either cause, reducing the exposure available for the other cause.

Multiple lives (2)

Joint life and last survivor annuities and insurances, `act_joint_life_last_survivor`

The joint life and last survivor annuities-due and whole life insurances (end of year of the first or second death) for two independent lives aged x and y with their own life tables.

$$\ddot{a}_{xy} = \sum_{k \geq 0} v^k \, {}_kp_x \, {}_kp_y, \quad \ddot{a}_{\overline{xy}} = \ddot{a}_x + \ddot{a}_y - \ddot{a}_{xy}, \quad A = 1 - d\,\ddot{a}$$

symbol		meaning	SI unit
`qx`	in	death probabilities of the first life from age x, ending with 1 (a list)	,
`qy`	in	death probabilities of the second life from age y, ending with 1 (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`a_joint`	out	joint life annuity-due ä_xy	,
`a_last`	out	last survivor annuity-due	,
`A_joint`	out	joint life insurance A_xy	,
`A_last`	out	last survivor insurance	,

Applies when: both tables close; independent lives; 0 < i < 1.

```
$ actuarial-science solve act_joint_life_last_survivor "qx=[0.1, 0.2, 0.5, 1]" "qy=[0.05,
0.1, 0.3, 0.6, 1]" i=0.05
a_joint = 2.55877551
a_last = 3.656643991
A_joint = 0.8781535471
A_last = 0.8258740957
  by: Joint life and last survivor annuities and insurances  (ä_xy = sum v^k k_p_x k_p_y;
last survivor ä = ä_x + ä_y - ä_xy; A = 1 - d ä)
  i = 0.05
  assumes: both tables close; independent lives; 0 < i < 1
```

What the worked example means: This calculation determines the present value of payments for two independent lives, aged x and y. The joint life annuity due, valued at 2.55878, pays only while both individuals are alive. Conversely, the last survivor annuity due, which contributes to the final result, ensures payments continue until the second death occurs. Using an annual effective interest rate of 0.05, the model computes the corresponding whole life insurance values, such as the joint life insurance of 0.878154, reflecting the cost of coverage that ends upon the first death.

Reversionary annuity, `act_reversionary_annuity`

The actuarial present value of an annuity-immediate of P a year paid to a life aged y after the death of a life aged x (a_(x|y) = a_y - a_xy), independent lives.

$$a_{x|y} = a_y - a_{xy} = \sum_{k\geq 1} v^k \, {}_kp_y \, (1 - {}_kp_x)$$

symbol		meaning	SI unit
`qx`	in	death probabilities of the life whose death starts the annuity, ending with 1 (a list)	,
`qy`	in	death probabilities of the annuitant from age y, ending with 1 (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`P`	in	annual payment	,
`a_rev`	out	reversionary annuity a_(x|y) per unit	,
`EPV`	out	actuarial present value P a_(x|y)	,

Applies when: both tables close; independent lives; 0 < i < 1.

```
$ actuarial-science solve act_reversionary_annuity "qx=[0.1, 0.2, 0.5, 1]" "qy=[0.05, 0.1,
0.3, 0.6, 1]" i=0.05 P=1000
a_rev = 0.8354583738
EPV = 835.4583738
  by: Reversionary annuity  (a_(x|y) = a_y - a_xy = sum_(k>=1) v^k k_p_y (1 - k_p_x))
  i = 0.05
  P = 1000
  assumes: both tables close; independent lives; 0 < i < 1
```

What the worked example means: This calculation determines the present worth of payments made to a life aged y after the death of a life aged x. It uses independent mortality probabilities for both lives and an interest rate of 0.05. The result, 835.458, represents the total expected cost today for an annual payment of 1000, based on the per-unit value 0.835458.

Premiums (6)

Gross annual premium for whole life insurance with expenses, `act_gross_premium`

The level annual gross premium for a fully discrete whole life insurance of S by the equivalence principle with expenses: an initial fraction e0 of the first premium, a renewal fraction r of later premiums, a per-policy expense c every year (first year included) and a settlement expense E per claim; with the net premium and the loading.

$$G = \frac{(S + E)A_x + c\,\ddot{a}_x}{(1 - r)\ddot{a}_x - (e_0 - r)}$$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... ending with 1 (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`S`	in	sum insured	,
`e0`	in	initial expense as a fraction of the first gross premium (e.g. 40%)	,
`r`	in	renewal expense as a fraction of each later gross premium (e.g. 5%)	,
`c`	in	per-policy expense at the start of every year	,

symbol		meaning	SI unit
`E`	in	settlement expense per claim	,
`G`	out	annual gross premium	,
`P`	out	annual net premium	,
`loading`	out	expense loading G - P	,

Applies when: the table closes; 0 < i < 1; 0 <= e0, r < 1; c, E >= 0.

```
$ actuarial-science solve act_gross_premium "q=[0.1, 0.2, 0.5, 1]" i=0.05 S=1000 e0=0.4
r=0.05 c=10 E=20
G = 391.0443937
P = 306.8418549
loading = 84.20253884
  by: Gross annual premium for whole life insurance with expenses  (G ä_x = (S + E) A_x +
e0 G + r G (ä_x - 1) + c ä_x)
  i = 0.05
  S = 1000
  e0 = 0.4
  r = 0.05
  c = 10
  E = 20
  assumes: the table closes; 0 < i < 1; 0 <= e0, r < 1; c, E >= 0
```

What the worked example means: This calculation determines the level annual gross premium required to fund a whole life insurance policy using the equivalence principle, which balances the present value of premiums against the present value of benefits and expenses. The result, 391.044, includes the pure risk cost represented by the net premium 306.842 plus the loading 84.2025 that covers administrative costs, commissions, and settlement fees. The inputs driving this figure include the sum insured 1000, the interest rate 0.05, and various expense parameters such as the initial expense fraction 0.4, renewal expense fraction 0.05, per-policy annual expense 10, and settlement expense 20.

Loss at issue of whole life insurance: mean, standard deviation, probability of loss, `act_loss_at_issue`

The mean and standard deviation of the insurer's loss at issue L0 = S v^(K+1) - P ä_(K+1) for a fully discrete whole life insurance of S with annual premium P, and the probability that the loss is positive.

$$L_0 = \left(S + \frac{P}{d}\right) v^{K+1} - \frac{P}{d}, \quad \mathrm{Var}(L_0) = \left(S + \frac{P}{d}\right)^2 \left({}^2A_x - A_x^2\right)$$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... ending with 1 (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`S`	in	sum insured	,
`P`	in	annual premium	,
`mean`	out	expected loss at issue S A_x - P ä_x	,
`sd`	out	standard deviation (S + P/d) sqrt(2A_x - A_x^2)	,

symbol		meaning	SI unit
`prob_loss`	out	probability that the loss at issue is positive	,

Applies when: the table closes; 0 < i < 1; S, P > 0.

```
$ actuarial-science solve act_loss_at_issue "q=[0.1, 0.2, 0.5, 1]" i=0.05 S=1000 P=450
mean = -403.8756999
sd = 434.1005558
prob_loss = 0.28
  by: Loss at issue of whole life insurance: mean, standard deviation, probability of loss
(L0 = (S + P/d) v^(K+1) - P/d; E = S A - P ä; Var = (S + P/d)^2 (2A - A^2))
  i = 0.05
  S = 1000
  P = 450
  assumes: the table closes; 0 < i < 1; S, P > 0
```

What the worked example means: This calculation determines the insurer's financial risk for a fully discrete whole life insurance policy. It computes the mean loss, which equals the present value of the sum insured minus the present value of premiums, resulting in -403.876. The standard deviation measures the variability of this loss around the mean, calculated as 434.101. Additionally, it provides the probability that the actual loss will be positive, indicating the likelihood that claims exceed collected premiums, which is 0.28. These metrics rely on the given death probabilities, interest rate, sum insured, and annual premium.

Net annual premium for endowment insurance, `act_net_premium_endowment`

The level annual net premium, payable for n years while alive, for a fully discrete n-year endowment insurance of S, by the equivalence principle.

$$P = S\,\frac{A_{x:\overline{n}|}}{\ddot{a}_{x:\overline{n}|}} = S\left(\frac{1}{\ddot{a}_{x:\overline{n}|}} - d\right)$$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... for at least n years (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`n`	in	term in years	,
`S`	in	sum insured	,
`P`	out	annual net premium S A_(x:n) / ä_(x:n)	,
`A`	out	endowment insurance A_(x:n)	,
`a_due`	out	temporary annuity-due ä_(x:n)	,

Applies when: at least n death probabilities; 0 < i < 1; S > 0.

```
$ actuarial-science solve act_net_premium_endowment "q=[0.1, 0.2, 0.5, 1]" i=0.05 n=3
S=1000
P = 350.7549361
A = 0.8804664723
a_due = 2.510204082
  by: Net annual premium for endowment insurance  (P ä_(x:n) = S A_(x:n); equivalently P =
S (1/ä_(x:n) - d))
  i = 0.05
  S = 1000
  assumes: at least n death probabilities; 0 < i < 1; S > 0
```

What the worked example means: This calculation determines the level annual net premium, payable for n years while alive, for a fully discrete n-year endowment insurance. Using the equivalence principle, the result is 350.755, which balances the present value of benefits against the present value of premiums. The underlying components are the endowment insurance factor 0.880466 and the temporary life annuity-due factor 2.5102, both computed from the provided death probabilities, interest rate, term, and sum insured.

Net annual premium for limited-payment whole life insurance, `act_net_premium_limited_pay`

The level annual net premium for a fully discrete whole life insurance of S with premiums payable for at most h years.

$$ {}_hP_x = S\,\frac{A_x}{\ddot{a}_{x:\overline{h}|}} $$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... ending with 1 (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`h`	in	premium paying period in years	,
`S`	in	sum insured	,
`P`	out	annual net premium S A_x / ä_(x:h)	,
`A`	out	whole life insurance A_x	,
`a_due_h`	out	h-year temporary annuity-due	,

Applies when: the table closes; 1 <= h <= length of the table; 0 < i < 1.

```
$ actuarial-science solve act_net_premium_limited_pay "q=[0.1, 0.2, 0.5, 1]" i=0.05 h=2
S=1000
P = 466.1234457
A = 0.8656578278
a_due_h = 1.857142857
  by: Net annual premium for limited-payment whole life insurance  (P ä_(x:h) = S A_x)
  i = 0.05
  S = 1000
  assumes: the table closes; 1 <= h <= length of the table; 0 < i < 1
```

What the worked example means: This calculation determines the level annual net premium required for a fully discrete whole life insurance policy with a sum insured of 1000. The premium is payable for at most 2 years. It is derived by dividing the present value of the death benefit, represented by the whole life insurance

factor 0.865658, by the present value of the premium payments, represented by the temporary annuity-due factor 1.85714. The resulting annual payment amount is 466.123. This ensures that the expected present value of premiums equals the expected present value of benefits under the given interest rate of 0.05.

Net annual premium for term insurance, `act_net_premium_term`

The level annual net premium, payable for n years while alive, for a fully discrete n-year term insurance of S, by the equivalence principle.

$$P = S \frac{A^1_{x:\overline{n}|}}{\ddot{a}_{x:\overline{n}|}}$$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... for at least n years (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`n`	in	term in years	,
`S`	in	sum insured	,
`P`	out	annual net premium S A^1_(x:n) / ä_(x:n)	,
`A1`	out	term insurance A^1_(x:n)	,
`a_due`	out	temporary annuity-due ä_(x:n)	,

Applies when: at least n death probabilities; 0 < i < 1; S > 0.

```
$ actuarial-science solve act_net_premium_term "q=[0.1, 0.2, 0.5, 1]" i=0.05 n=3 S=1000
P = 226.867983
A1 = 0.5694849368
a_due = 2.510204082
  by: Net annual premium for term insurance  (P ä_(x:n) = S A^1_(x:n))
  i = 0.05
  S = 1000
  assumes: at least n death probabilities; 0 < i < 1; S > 0
```

What the worked example means: This calculation determines the level annual net premium for a fully discrete term insurance policy using the equivalence principle. The result, 226.868, represents the amount paid each year while the insured is alive for the term duration. It is computed by dividing the present value of the death benefit, denoted as 0.569485, by the present value of the temporary life annuity-due, denoted as 2.5102. These intermediate values reflect the actuarial present values based on the provided mortality probabilities and interest rate.

Net annual premium for whole life insurance, `act_net_premium_whole_life`

The level annual net premium, payable at the start of each year while alive, for a fully discrete whole life insurance of S, by the equivalence principle.

$$P = S \frac{A_x}{\ddot{a}_x} = S\left(\frac{1}{\ddot{a}_x} - d\right)$$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... ending with 1 (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`S`	in	sum insured	,
`P`	out	annual net premium S A_x / ä_x	,
`A`	out	whole life insurance A_x	,
`a_due`	out	whole life annuity-due ä_x	,

Applies when: the table closes; 0 < i < 1; S > 0.

```
$ actuarial-science solve act_net_premium_whole_life "q=[0.1, 0.2, 0.5, 1]" i=0.05 S=1000
P = 306.8418549
A = 0.8656578278
a_due = 2.821185617
  by: Net annual premium for whole life insurance  (equivalence principle: P ä_x = S A_x)
  i = 0.05
  S = 1000
  assumes: the table closes; 0 < i < 1; S > 0
```

What the worked example means: This calculation determines the level annual net premium, payable at the start of each year while the insured is alive, for a fully discrete whole life insurance policy. Using the equivalence principle, the result 306.842 balances the present value of benefits against the present value of premiums. The inputs include a death probability list ending in one, an interest rate of 0.05, and a sum insured of 1000. The intermediate values are the whole life insurance factor 0.865658 and the annuity-due factor 2.82119. A higher interest rate reduces both the insurance value and the required premium, while a larger sum insured directly increases the premium without changing the underlying factors.

Ratemaking (4)

Indicated rate by the pure premium method, `act_pure_premium_rate`

The pure premium (losses per exposure) and the indicated rate per exposure from projected losses, exposures, the fixed expense per exposure, the variable expense ratio and the profit provision.

$$R = \frac{L/X + F}{1 - V - Q}$$

symbol		meaning	SI unit
`L`	in	projected ultimate losses and LAE	,
`X`	in	earned exposures	,
`Fe`	in	fixed expense per exposure	,
`V`	in	variable expense ratio	,
`Q`	in	profit and contingencies provision	,
`pure_premium`	out	pure premium L/X	,
`rate`	out	indicated rate per exposure (PP + F)/(1 - V - Q)	,

Applies when: X > 0; 0 <= V < 1; V + Q < 1.

```
$ actuarial-science solve act_pure_premium_rate L=500000 X=2000 Fe=30 V=0.15 Q=0.05
pure_premium = 250
rate = 350
  by: Indicated rate by the pure premium method  (rate = (L/X + F)/(1 - V - Q))
  L = 500000
  X = 2000
  Fe = 30
  V = 0.15
  Q = 0.05
  assumes: X > 0; 0 <= V < 1; V + Q < 1
```

What the worked example means: This calculation determines the indicated rate per exposure using the pure premium method. It combines projected ultimate losses and LAE with earned exposures to find the pure premium, then adds fixed expenses per exposure. This total cost is divided by one minus the variable expense ratio and the profit provision to yield the final indicated rate of 350, based on inputs where exposures are positive and expense plus profit ratios remain below one.

Indicated rate change by the loss ratio method, `act_loss_ratio_rate_change`

The indicated rate change from projected losses (with loss adjustment expenses), earned premium at current rates, the fixed expense ratio, the variable expense ratio and the profit provision.

$$\Delta = \frac{L/EP + F}{1 - V - Q} - 1$$

symbol		meaning	SI unit
`L`	in	projected ultimate losses and LAE	,
`EP`	in	earned premium at current rates	,
`F`	in	fixed expense ratio (to premium)	,
`V`	in	variable expense ratio	,
`Q`	in	profit and contingencies provision	,
`change`	out	indicated rate change (0.05 = +5%)	,
`loss_ratio`	out	projected loss ratio L/EP	,
`permissible`	out	permissible loss ratio 1 - V - Q	,

Applies when: EP > 0; 0 <= F, V < 1; V + Q < 1.

```
$ actuarial-science solve act_loss_ratio_rate_change L=650000 EP=1000000 F=0.1 V=0.15
Q=0.05
change = -0.0625
loss_ratio = 0.65
permissible = 0.8
  by: Indicated rate change by the loss ratio method  (indicated change = (L/EP + F)/(1 - V
- Q) - 1)
  L = 650000
  EP = 1000000
  F = 0.1
  V = 0.15
  Q = 0.05
  assumes: EP > 0; 0 <= F, V < 1; V + Q < 1
```

What the worked example means: This calculation determines the indicated rate change by comparing the projected ultimate losses and LAE against earned premium at current rates. The result, -0.0625, reflects the adjustment needed to align the actual loss experience with the target permissible loss ratio. This process incorporates fixed expense ratios, variable expense ratios, and profit provisions to ensure the new rates cover all expected costs and desired margins.

Loss and premium trend factors to a future period, `act_trend_projection`

The trend period of each experience year (from its midpoint to the average date of the future period), the compound trend factor at an annual trend rate, and the trended losses or premiums.

$$t_y = D - (y + \tfrac{1}{2}), \qquad \text{trended}_y = X_y\,(1+g)^{t_y}$$

symbol		meaning	SI unit
`amounts`	in	losses (or premiums) of each experience year (a list)	,
`years`	in	experience years (whole numbers, a list)	,
`trend`	in	annual trend rate (3% or 0.03; negative for a falling trend)	,
`to_date`	in	average date of the future period as a decimal year (2025.5 = 1 July 2025)	,
`periods`	out	trend period of each year in years: to_date - (year + 0.5) (a list)	,
`factors`	out	trend factor (1 + trend)^period of each year (a list)	,
`trended`	out	trended amount of each year (a list)	,
`total_trended`	out	total of the trended amounts	,

Applies when: one year per amount; whole years; amounts >= 0; -100% < trend < 100%; to_date >= year + 0.5.

```
$ actuarial-science solve act_trend_projection "amounts=[100000, 120000]" "years=[2022,
2023]" trend=0.05 to_date=2025.5
periods = [3.0,2.0]
factors = [1.157625,1.1025]
trended = [115762.5,132300.0]
total_trended = 248062.5
  by: Loss and premium trend factors to a future period  (period = to_date - (year + 0.5);
factor = (1 + trend)^period; trended = amount x factor)
  trend = 0.05
  to_date = 2025.5
  assumes: one year per amount; whole years; amounts >= 0; -100% < trend < 100%; to_date >=
year + 0.5
```

In the worked example, amounts = [100000, 120000], years = [2022, 2023], trend = 0.05, to_date = 2025.5 go through the equation above and give periods = [3, 2], factors = [1.157625, 1.1025], trended = [115762.5, 132300], total_trended = 248062.5. It applies only when one year per amount; whole years; amounts >= 0; -100% < trend < 100%; to_date >= year + 0.5: outside that it says what is wrong instead of returning a number.

On-level earned premium by the parallelogram method, `act_on_level_premium`

The average rate level of each calendar year's earned premium under past rate changes (policies written evenly, earned evenly over their term), the on-level factor that brings it to the current rate level, and the earned premium at current rates.

$$\bar{L}_Y = \sum_j \big(A(t_{j+1} - Y) - A(t_j - Y)\big) L_j, \qquad \text{factor}_Y = \frac{L_{\text{current}}}{\bar{L}_Y}$$

symbol		meaning	SI unit
`dates`	in	effective dates of the rate changes as decimal years, increasing (2022.5 = 1 July 2022)	,
`changes`	in	rate change at each date as a decimal (0.05 = +5%, -0.03 = -3%)	,
`years`	in	calendar years of the earned premium (whole numbers, increasing)	,
`EP`	in	earned premium of each calendar year (at the rates then charged)	,
`T`	in	policy term in years, at most 1 (1 = annual, 0.5 = six months)	,
`average_level`	out	average rate level of each calendar year's earned premium (a list)	,
`factors`	out	on-level factor of each year: current level / average level (a list)	,
`on_level_EP`	out	earned premium at current rate level of each year (a list)	,
`total_on_level_EP`	out	total earned premium at current rate level	,
`current_level`	out	current cumulative rate level (1 before the first change)	,

Applies when: one change per date, dates increasing; -100% < change < 100%; whole calendar years, increasing; earned premiums >= 0; 0 < T <= 1.

```
$ actuarial-science solve act_on_level_premium "dates=[2022.5]" "changes=[0.1]" "years=
[2022, 2023, 2024]" "EP=[1000, 1100, 1200]" T=1
average_level = [1.0125,1.0875,1.1]
factors = [1.08641975309,1.01149425287,1.0]
on_level_EP = [1086.41975309,1112.64367816,1200.0]
total_on_level_EP = 3399.063431
current_level = 1.1
  by: On-level earned premium by the parallelogram method  (share of CY earned exposure
written before d: (d+T)^2/(2T), T/2 + d, 1 - (1-d)^2/(2T) on the three parts of the
parallelogram; average level = sum of shares x levels; factor = current level / average
level)
  T = 1
  assumes: one change per date, dates increasing; -100% < change < 100%; whole calendar
years, increasing; earned premiums >= 0; 0 < T <= 1
```

In the worked example, dates = [2022.5], changes = [0.1], years = [2022, 2023, 2024], EP = [1000, 1100, 1200], T = 1 go through the equation above and give average_level = [1.0125, 1.0875, 1.1], factors = [1.086419753, 1.011494253, 1], on_level_EP = [1086.419753, 1112.643678, 1200], total_on_level_EP = 3399.063431. It applies only when one change per date, dates increasing; -100% < change < 100%; whole calendar years, increasing; earned premiums >= 0; 0 < T <= 1: outside that it says what is wrong instead of returning a number.

Reserves (8)

Continuous whole life insurance, premium and reserve under Makeham's law, `act_makeham_continuous_whole_life`

The continuous whole life annuity, the insurance paid at the moment of death, the continuous net premium rate and the reserve at time t for a life aged x following Makeham's law, at a constant interest rate.

$$\bar{a}_x = \int_0^\infty e^{-\delta s}\,{}_sp_x\,ds, \quad \bar{A}_x = 1 - \delta\bar{a}_x, \quad {}_t\bar{V} = S\left(1 - \frac{\bar{a}_{x+t}}{\bar{a}_x}\right)$$

symbol		meaning	SI unit
`A`	in	Makeham's A (per year)	,
`B`	in	Makeham's B (per year)	,
`c`	in	Makeham's c (> 1)	,
`x`	in	age at issue (years)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`S`	in	sum insured	,
`t`	in	time since issue in years	,
`a_bar`	out	continuous whole life annuity ā_x	,
`A_bar`	out	insurance payable at death Ā_x	,
`P`	out	continuous annual net premium rate	,
`V`	out	reserve at time t	,

Applies when: Makeham parameters 0 <= A < 0.1, 0 < B < 0.1, 1 < c < 1.5; 0 <= x <= 120; 0 <= t <= 100; 0 < i < 1.

```
$ actuarial-science solve act_makeham_continuous_whole_life A=0.00022 B=2.7e-06 c=1.124
x=50 i=0.05 S=100000 t=10
a_bar = 16.52037321
A_bar = 0.1939682791
P = 1174.1156
V = 12836.47174
  by: Continuous whole life insurance, premium and reserve under Makeham's law  (ā_x =
integral of e^(-delta s) s_p_x (Gauss-Legendre); Ā_x = 1 - delta ā_x; P = S Ā_x/ā_x; tV = S
(1 - ā_(x+t)/ā_x))
  A = 0.00022
  B = 2.7e-06
  c = 1.124
  x = 50
  i = 0.05
  S = 100000
  t = 10
  assumes: Makeham parameters 0 <= A < 0.1, 0 < B < 0.1, 1 < c < 1.5; 0 <= x <= 120; 0 <= t
<= 100; 0 < i < 1
```

What the worked example means: This calculation determines the financial values for a continuous whole life insurance policy issued to a person aged 50 years. It computes the actuarial present value of a continuous annuity, denoted as 16.5204, using Makeham's law parameters A, B, and c with an annual effective interest rate of 0.05. From this, it derives the net single premium for the insurance benefit payable at the moment of death, 0.193968, and the level annual continuous net premium rate, P. Finally, it calculates the prospective reserve at duration 10 years, which is 12836.5, based on a sum insured of 100000. These figures represent the expected cost and liability associated with the policy under the specified mortality and interest assumptions.

Gross premium reserve for whole life insurance with expenses, `act_gross_premium_reserve`

The gross premium policy value at integer duration t of a fully discrete whole life insurance of S, with the gross premium set by the equivalence principle with initial, renewal, per-policy and settlement expenses.

$$ {}_tV^g = (S+E)A_{x+t} + c\,\ddot{a}_{x+t} - (1-r)G\,\ddot{a}_{x+t} $$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... ending with 1 (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`S`	in	sum insured	,
`e0`	in	initial expense as a fraction of the first gross premium	,
`r`	in	renewal expense as a fraction of each later gross premium	,
`c`	in	per-policy expense at the start of every year	,
`E`	in	settlement expense per claim	,
`t`	in	policy duration in years	,
`V`	out	gross premium reserve tV	,

symbol		meaning	SI unit
`G`	out	annual gross premium	,

Applies when: the table closes; 0 <= t < length of the table; 0 <= e0, r < 1.

```
$ actuarial-science solve act_gross_premium_reserve "q=[0.1, 0.2, 0.5, 1]" i=0.05 S=1000
e0=0.4 r=0.05 c=10 E=20 t=1
V = 148.7310756
G = 391.0443937
  by: Gross premium reserve for whole life insurance with expenses  (tV = (S + E) A_(x+t) +
c ä_(x+t) - G (1 - r) ä_(x+t) for t >= 1 (0 at issue by the equivalence principle))
  i = 0.05
  S = 1000
  e0 = 0.4
  r = 0.05
  c = 10
  E = 20
  assumes: the table closes; 0 <= t < length of the table; 0 <= e0, r < 1
```

What the worked example means: This calculation determines the gross premium policy value at duration t, representing the insurer's liability for a whole life insurance contract. It accounts for future benefits and expenses against expected future premiums. The result, 148.731, is computed using an annual gross premium of 391.044, which was set by the equivalence principle to cover initial costs of 0.4 fraction, renewal costs of 0.05 fraction, per-policy expenses of 10, and settlement expenses of 20. This reserve reflects the present value of all future outgo minus future income under the specified interest rate of 0.05.

Net premium reserve for endowment insurance, `act_reserve_endowment`

The net premium reserve at integer duration t of a fully discrete n-year endowment insurance of S with level net premiums payable for n years, by the prospective method (S at maturity).

$$ {}_tV = S\,A_{x+t:\overline{n-t}|} - P\,\ddot{a}_{x+t:\overline{n-t}|} $$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... for at least n years (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`n`	in	term in years	,
`S`	in	sum insured	,
`t`	in	policy duration in years	,
`V`	out	net premium reserve tV	,
`P`	out	annual net premium	,

Applies when: at least n death probabilities; 0 <= t <= n; 0 < i < 1.

```
$ actuarial-science solve act_reserve_endowment "q=[0.1, 0.2, 0.5, 1]" i=0.05 n=3 S=1000
t=1
V = 298.102981
P = 350.7549361
  by: Net premium reserve for endowment insurance  (tV = S A_(x+t:n-t) - P ä_(x+t:n-t))
  i = 0.05
  S = 1000
  assumes: at least n death probabilities; 0 <= t <= n; 0 < i < 1
```

What the worked example means: This calculation determines the net premium reserve at duration t for an n-year endowment insurance policy. Using the prospective method, the reserve equals the present value of future benefits minus the present value of future premiums. With a sum insured of 1000, term of 3 years, interest rate of 0.05, and current duration of 1, the resulting annual net premium is 350.755 and the reserve held at time t is 298.103. This value represents the liability the insurer must hold to cover future obligations under the policy assumptions.

Net premium reserve for term insurance, `act_reserve_term`

The net premium reserve at integer duration t of a fully discrete n-year term insurance of S with level net premiums payable for n years, by the prospective method.

$$ {}_tV = S\,A^{1}_{x+t:\overline{n-t}|} - P\,\ddot{a}_{x+t:\overline{n-t}|} $$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... for at least n years (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`n`	in	term in years	,
`S`	in	sum insured	,
`t`	in	policy duration in years	,
`V`	out	net premium reserve tV	,
`P`	out	annual net premium	,

Applies when: at least n death probabilities; 0 <= t <= n; 0 < i < 1.

```
$ actuarial-science solve act_reserve_term "q=[0.01, 0.02, 0.04, 0.08]" i=0.05 n=4 S=1000
t=2
V = 43.21035282
P = 33.71740631
  by: Net premium reserve for term insurance  (tV = S A^1_(x+t:n-t) - P ä_(x+t:n-t))
  i = 0.05
  S = 1000
  assumes: at least n death probabilities; 0 <= t <= n; 0 < i < 1
```

What the worked example means: This calculation determines the net premium reserve at duration t for a term insurance policy. It uses the prospective method, subtracting the present value of future premiums from the present value of future benefits. The result, 43.2104, represents the funds held to cover future obligations given the interest rate 0.05, term 4, and sum insured 1000. The annual net premium used in this calculation is 33.7174.

Net premium reserve for whole life insurance, `act_reserve_whole_life`

The net premium reserve (policy value) at integer duration t of a fully discrete whole life insurance of S with the equivalence-principle net premium, by the prospective method.

$$ {}_tV = S\,A_{x+t} - P\,\ddot{a}_{x+t} = S\left(1 - \frac{\ddot{a}_{x+t}}{\ddot{a}_x}\right) $$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... ending with 1 (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`S`	in	sum insured	,
`t`	in	policy duration in years	,
`V`	out	net premium reserve tV	,
`P`	out	annual net premium	,
`A_t`	out	A_(x+t)	,
`a_due_t`	out	ä_(x+t)	,

Applies when: the table closes; 0 <= t < length of the table; 0 < i < 1.

```
$ actuarial-science solve act_reserve_whole_life "q=[0.1, 0.2, 0.5, 1]" i=0.05 S=1000 t=1
V = 246.8710529
P = 306.8418549
A_t = 0.8988230213
a_due_t = 2.124716553
  by: Net premium reserve for whole life insurance  (tV = S A_(x+t) - P ä_(x+t) = S (1 -
ä_(x+t)/ä_x))
  i = 0.05
  S = 1000
  assumes: the table closes; 0 <= t < length of the table; 0 < i < 1
```

What the worked example means: This calculation determines the prospective net premium reserve at duration t for a whole life insurance policy. It subtracts the present value of future premiums, represented by 306.842 multiplied by the annuity due factor 2.12472, from the present value of future benefits, which is the sum insured 1000 times the insurance factor 0.898823. The resulting reserve 246.871 indicates the liability the insurer must hold to cover future obligations under the equivalence principle.

Policy value recursion (one year), `act_policy_value_recursion`

The policy value at the end of a policy year from the value at its start, the premium and expense at the start of the year, the interest rate, the death probability, the sum insured and the settlement expense: (tV + P - e)(1+i) = q (S + E) + p (t+1)V.

$$ ({}_tV + P - e)(1+i) = q_{x+t}\,(S+E) + p_{x+t}\;{}_{t+1}V $$

symbol		meaning	SI unit
`V`	in	policy value at the start of the year tV	,

symbol		meaning	SI unit
`P`	in	premium at the start of the year	,
`e`	in	expense at the start of the year	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`q`	in	death probability for the year q_(x+t)	,
`S`	in	death benefit	,
`E`	in	settlement expense per claim	,
`V_next`	out	policy value at the end of the year (t+1)V	,

Applies when: 0 <= q < 1; 0 < i < 1.

```
$ actuarial-science solve act_policy_value_recursion V=100 P=50 e=5 i=0.05 q=0.01 S=1000
E=0
V_next = 143.6868687
  by: Policy value recursion (one year)  ((t+1)V = [(tV + P - e)(1 + i) - q (S + E)] / (1 -
q))
  V = 100
  P = 50
  e = 5
  i = 0.05
  q = 0.01
  S = 1000
  E = 0
  assumes: 0 <= q < 1; 0 < i < 1
```

What the worked example means: This calculation determines the prospective policy reserve at the end of the current year. It begins with the reserve held at the start of the year, adds the premium received, and subtracts initial expenses. This net amount accumulates with interest over the year. From this accumulated fund, the expected cost of claims (death benefit plus settlement expenses weighted by the probability of death) is deducted. The remaining amount is divided by the probability of survival to determine the required reserve for those who survive to the next year. The resulting value is 143.687.

Retrospective reserve (accumulated fund per survivor), `act_retrospective_reserve`

The retrospective reserve at integer duration t of a fully discrete life insurance of S with annual premium P: the actuarial accumulated value of the premiums received less the cost of insurance, per survivor.

$$ {}_tV = \frac{P\,\ddot{a}_{x:\overline{t}|} - S\,A^{1}_{x:\overline{t}|}}{{}_tE_x} $$

symbol		meaning	SI unit
`q`	in	death probabilities q_x, q_(x+1), ... for at least t years (a list)	,
`i`	in	annual effective interest rate (5% or 0.05)	,
`S`	in	sum insured	,
`P`	in	annual premium	,
`t`	in	policy duration in years	,

symbol		meaning	SI unit
`V`	out	retrospective reserve tV	,
`tEx`	out	pure endowment t_E_x	,

Applies when: at least t death probabilities; t_p_x > 0; 0 < i < 1.

```
$ actuarial-science solve act_retrospective_reserve "q=[0.1, 0.2, 0.5, 1]" i=0.05 S=1000
P=450 t=2
V = 883.8541667
tEx = 0.6530612245
  by: Retrospective reserve (accumulated fund per survivor)  (tV = (P ä_(x:t) - S
A^1_(x:t)) / t_E_x)
  i = 0.05
  S = 1000
  P = 450
  assumes: at least t death probabilities; t_p_x > 0; 0 < i < 1
```

What the worked example means: This calculation determines the retrospective reserve, which equals the actuarial accumulated value of premiums received minus the cost of insurance, divided by the pure endowment factor. Using a sum insured of 1000 and an annual premium of 450, the resulting reserve per survivor is 883.854. The pure endowment factor used in this division is 0.653061, reflecting survival probability and interest accumulation over 2 years.

Thiele's equation: continuous endowment insurance reserve (constant forces), `act_thiele_endowment`

The continuous net premium rate and the reserve at time t of an n-year endowment insurance of S paid at the moment of death or at maturity, with a constant force of mortality mu and a constant force of interest delta (the solution of Thiele's equation dV/dt = delta V + P - mu (S - V), V(n) = S).

$$\frac{d\,{}_tV}{dt} = \delta\,{}_tV + \bar{P} - \mu\,(S - {}_tV), \quad {}_nV = S$$

symbol		meaning	SI unit
`mu`	in	constant force of mortality per year	,
`delta`	in	constant force of interest per year	,
`n`	in	term in years	,
`S`	in	sum insured	,
`t`	in	time since issue in years	,
`P`	out	continuous annual net premium rate	,
`A_bar`	out	endowment insurance Ā_(x:n)	,
`a_bar`	out	continuous temporary annuity ā_(x:n)	,
`V`	out	reserve at time t	,

Applies when: 0 < mu, delta < 1; 0 < n <= 100; 0 <= t <= n.

```
$ actuarial-science solve act_thiele_endowment mu=0.02 delta=0.05 n=20 S=1000 t=10
P = 42.91175725
A_bar = 0.4618549742
a_bar = 10.76290052
V = 331.8122278
  by: Thiele's equation: continuous endowment insurance reserve (constant forces)  (Ā =
mu/(mu+delta)(1 - e^(-(mu+delta)n)) + e^(-(mu+delta)n); ā = (1 - e^(-
(mu+delta)n))/(mu+delta); P = S Ā/ā; tV = S Ā_(n-t) - P ā_(n-t))
  mu = 0.02
  delta = 0.05
  n = 20
  S = 1000
  t = 10
  assumes: 0 < mu, delta < 1; 0 < n <= 100; 0 <= t <= n
```

What the worked example means: This calculation determines the financial reserves for an endowment insurance policy using Thiele's differential equation. Given a constant force of mortality of 0.02 and a constant force of interest of 0.05, it first computes the present value of benefits, denoted as 0.461855, and the present value of premiums, denoted as 10.7629. These values give the continuous annual net premium rate, which is 42.9118. Finally, applying these rates over the term of 20 years, the policy reserve accumulated at time 10 is 331.812. This reserve represents the funds held by the insurer to meet future obligations for this specific policy.

Risk measures (4)

VaR and TVaR of a lognormal loss, `act_var_tvar_lognormal`

The value at risk and tail value at risk at security level p of a lognormal loss (mu, sigma of the log of the loss).

$$\mathrm{VaR}_p = e^{\mu+\sigma z_p}, \quad \mathrm{TVaR}_p = e^{\mu+\sigma^2/2}\,\frac{\Phi(\sigma - z_p)}{1-p}$$

symbol		meaning	SI unit
`mu`	in	mean of the log of the loss	,
`sigma`	in	standard deviation of the log of the loss	,
`p`	in	security level (95% or 0.95)	,
`VaR`	out	value at risk at level p (the p-quantile)	,
`TVaR`	out	tail value at risk E[X | X > VaR]	,

Applies when: sigma > 0; 0 < p < 1.

```
$ actuarial-science solve act_var_tvar_lognormal mu=7 sigma=1 p=0.95
VaR = 5680.835676
TVaR = 9384.138726
  by: VaR and TVaR of a lognormal loss  (VaR = exp(mu + sigma z_p); TVaR = e^(mu +
sigma^2/2) Phi(sigma - z_p)/(1 - p))
  mu = 7
  sigma = 1
  p = 0.95
  assumes: sigma > 0; 0 < p < 1
```

What the worked example means: This calculation determines the Value at Risk and Tail Value at Risk for a lognormal loss distribution. Given the mean of the log loss as 7, the standard deviation of the log loss as 1, and a security level of 0.95, the model computes the p-quantile of the loss. The resulting Value at Risk is 5680.84, representing the threshold loss exceeded with probability one minus p. The Tail Value at Risk is 9384.14, which represents the expected loss amount given that the loss exceeds this threshold.

VaR and TVaR of a normal loss, `act_var_tvar_normal`

The value at risk and tail value at risk at security level p of a normally distributed loss.

$$\mathrm{VaR}_p = \mu + \sigma z_p, \quad \mathrm{TVaR}_p = \mu + \sigma \frac{\phi(z_p)}{1-p}$$

symbol		meaning	SI unit
`mu`	in	mean loss	,
`sigma`	in	standard deviation of the loss	,
`p`	in	security level (95% or 0.95)	,
`VaR`	out	value at risk at level p (the p-quantile)	,
`TVaR`	out	tail value at risk E[X | X > VaR]	,

Applies when: sigma > 0; 0 < p < 1.

```
$ actuarial-science solve act_var_tvar_normal mu=1000 sigma=200 p=0.95
VaR = 1328.970725
TVaR = 1412.542561
  by: VaR and TVaR of a normal loss  (VaR = mu + sigma z_p; TVaR = mu + sigma phi(z_p)/(1 -
p))
  mu = 1000
  sigma = 200
  p = 0.95
  assumes: sigma > 0; 0 < p < 1
```

What the worked example means: This calculation determines the Value at Risk and Tail Value at Risk for a loss distribution assumed to be normal. Using the mean loss of 1000, standard deviation of 200, and security level of 0.95, the model computes the p-quantile as 1328.97. It also calculates the expected loss given that the loss exceeds this quantile, resulting in 1412.54. These metrics indicate the threshold for extreme losses and the average severity beyond that threshold.

VaR and TVaR of a Pareto loss, `act_var_tvar_pareto`

The value at risk and tail value at risk at security level p of a two-parameter Pareto loss (alpha, theta).

$$\mathrm{VaR}_p = \theta\left[(1-p)^{-1/\alpha} - 1\right], \quad \mathrm{TVaR}_p = \mathrm{VaR}_p + \frac{\theta + \mathrm{VaR}_p}{\alpha - 1}$$

symbol		meaning	SI unit
`alpha`	in	Pareto shape alpha (> 1)	,

symbol		meaning	SI unit
`theta`	in	Pareto scale theta	,
`p`	in	security level (95% or 0.95)	,
`VaR`	out	value at risk at level p (the p-quantile)	,
`TVaR`	out	tail value at risk E[X | X > VaR]	,

Applies when: alpha > 1; theta > 0; 0 < p < 1.

```
$ actuarial-science solve act_var_tvar_pareto alpha=3 theta=2000 p=0.95
VaR = 3428.835233
TVaR = 6143.25285
  by: VaR and TVaR of a Pareto loss  (VaR = theta [(1-p)^(-1/alpha) - 1]; TVaR = VaR +
(theta + VaR)/(alpha - 1))
  alpha = 3
  theta = 2000
  p = 0.95
  assumes: alpha > 1; theta > 0; 0 < p < 1
```

What the worked example means: This calculation determines the Value at Risk and Tail Value at Risk for a two-parameter Pareto distribution. Using the specified shape parameter 3, scale parameter 2000, and security level 0.95, the model computes the p-quantile of the loss distribution as 3428.84. It then calculates the expected excess loss beyond that quantile, yielding a Tail Value at Risk of 6143.25. These metrics quantify the potential severity of losses in the upper tail of the distribution, providing a measure of extreme financial risk exposure.

VaR and TVaR of an exponential loss, `act_var_tvar_exponential`

The value at risk and tail value at risk at security level p of an exponentially distributed loss with mean theta.

$$\mathrm{VaR}_p = -\theta \ln(1-p), \quad \mathrm{TVaR}_p = \mathrm{VaR}_p + \theta$$

symbol		meaning	SI unit
`theta`	in	mean loss theta	,
`p`	in	security level (95% or 0.95)	,
`VaR`	out	value at risk at level p (the p-quantile)	,
`TVaR`	out	tail value at risk E[X | X > VaR]	,

Applies when: theta > 0; 0 < p < 1.

```
$ actuarial-science solve act_var_tvar_exponential theta=1000 p=0.95
VaR = 2995.732274
TVaR = 3995.732274
  by: VaR and TVaR of an exponential loss  (VaR = -theta ln(1 - p); TVaR = VaR + theta)
  theta = 1000
  p = 0.95
  assumes: theta > 0; 0 < p < 1
```

What the worked example means: This calculation determines the Value at Risk and Tail Value at Risk for an exponential loss distribution. Given a mean loss of 1000 and a security level of 0.95, the Value at Risk represents the threshold that losses exceed with probability one minus p, calculated as 2995.73. The Tail Value at Risk, which is the expected loss given that the loss exceeds the Value at Risk, is 3995.73. These metrics help quantify potential extreme financial impacts under the assumed model.

Survival estimation (3)

Exponential maximum-likelihood fit with censoring, `act_exponential_mle`

The maximum-likelihood constant hazard (deaths divided by total time observed), the mean lifetime and the fitted survival probability at time t from times with death (1) or censoring (0) indicators.

$$\hat{\lambda} = \frac{\sum_i \delta_i}{\sum_i t_i}, \quad \hat{S}(t) = e^{-\hat{\lambda} t}$$

symbol		meaning	SI unit
`times`	in	observed times (a list)	,
`events`	in	1 for a death or failure, 0 for censoring (a list)	,
`t`	in	time for the fitted survival probability	,
`hazard`	out	maximum-likelihood hazard rate (deaths / total exposure)	,
`mean`	out	fitted mean lifetime	,
`S`	out	fitted survival probability exp(-lambda t)	,

Applies when: one indicator per time; positive times; at least one death.

```
$ actuarial-science solve act_exponential_mle "times=[1, 2, 2, 3, 5]" "events=[1, 1, 0, 1,
0]" t=3
hazard = 0.2307692308
mean = 4.333333333
S = 0.5004199204
  by: Exponential maximum-likelihood fit with censoring  (lambda = sum of death indicators
/ sum of times; S(t) = e^(-lambda t))
  t = 3
  assumes: one indicator per time; positive times; at least one death
```

What the worked example means: This calculation determines the maximum-likelihood estimate for a constant hazard rate by dividing the total number of observed deaths by the sum of all observed times. The resulting hazard rate, 0.230769, implies an expected mean lifetime of 4.33333. Using this estimated rate, the model computes the probability that an individual survives beyond time 3, which yields a value of 0.50042. This approach assumes that the risk of failure remains constant over the entire observation period.

Kaplan-Meier product-limit survival estimate, `act_kaplan_meier`

The Kaplan-Meier estimate of the survival function at time t from observed times with death (1) or censoring (0) indicators; a life censored at a death time is counted at risk then.

$$\hat{S}(t) = \prod_{t_j \le t} \left(1 - \frac{d_j}{n_j}\right)$$

symbol		meaning	SI unit
`times`	in	observed times (a list)	,
`events`	in	1 for a death or failure, 0 for censoring (a list)	,
`t`	in	time at which the survival is wanted	,
`S`	out	Kaplan-Meier survival estimate S(t)	,
`F`	out	estimated probability of death by t	,
`deaths`	out	number of deaths observed up to t	,

Applies when: one indicator per time; positive times; indicators 0 or 1.

```
$ actuarial-science solve act_kaplan_meier "times=[1, 2, 2, 3, 5]" "events=[1, 1, 0, 1, 0]"
t=3
S = 0.3
F = 0.7
deaths = 3
  by: Kaplan-Meier product-limit survival estimate  (S(t) = product over death times t_j <=
t of (1 - d_j/n_j))
  t = 3
  assumes: one indicator per time; positive times; indicators 0 or 1
```

What the worked example means: This calculation determines the Kaplan-Meier product-limit survival estimate for a specified time using observed event times and censoring indicators. The result, 0.3, represents the probability that a subject survives beyond the time 3, while 0.7 indicates the cumulative probability of experiencing the event by that same time. These values are derived from the sequence of deaths and censored observations, where each death reduces the survival probability based on the number of individuals still at risk.

Nelson-Aalen cumulative hazard estimate, `act_nelson_aalen`

The Nelson-Aalen estimate of the cumulative hazard at time t from observed times with death (1) or censoring (0) indicators, and the survival estimate exp(-H(t)).

$$\hat{H}(t) = \sum_{t_j \le t} \frac{d_j}{n_j}, \quad \hat{S}(t) = e^{-\hat{H}(t)}$$

symbol		meaning	SI unit
`times`	in	observed times (a list)	,
`events`	in	1 for a death or failure, 0 for censoring (a list)	,
`t`	in	time at which the estimate is wanted	,
`H`	out	Nelson-Aalen cumulative hazard H(t)	,
`S`	out	survival estimate exp(-H(t))	,

Applies when: one indicator per time; positive times; indicators 0 or 1.

```
$ actuarial-science solve act_nelson_aalen "times=[1, 2, 2, 3, 5]" "events=[1, 1, 0, 1, 0]"
t=3
H = 0.95
S = 0.3867410235
  by: Nelson-Aalen cumulative hazard estimate  (H(t) = sum over death times t_j <= t of
d_j/n_j)
  t = 3
  assumes: one indicator per time; positive times; indicators 0 or 1
```

What the worked example means: This calculation determines the Nelson-Aalen cumulative hazard estimate, denoted as 0.95, by summing the ratio of deaths to those at risk for all event times up to 3. The corresponding survival probability, 0.386741, is derived by taking the exponential of the negative cumulative hazard. These values summarize the failure experience observed in the dataset up to the specified time point.

Survival models (4)

Constant force of mortality (exponential lifetime), `act_constant_force_mortality`

Survival and death probabilities, the complete and curtate expectations of life and the variance of the future lifetime when the force of mortality mu is constant.

$$ {}_tp_x = e^{-\mu t}, \quad \mathring{e}_x = \frac{1}{\mu}, \quad e_x = \frac{e^{-\mu}}{1 - e^{-\mu}} $$

symbol		meaning	SI unit
`mu`	in	constant force of mortality per year	,
`t`	in	number of years	,
`tpx`	out	probability of surviving t years	,
`tqx`	out	probability of dying within t years	,
`e_complete`	out	complete expectation 1/mu	,
`e_curtate`	out	curtate expectation p/(1-p)	,
`var_T`	out	variance of the future lifetime 1/mu^2	,

Applies when: 0 < mu < 5; t >= 0.

```
$ actuarial-science solve act_constant_force_mortality mu=0.02 t=10
tpx = 0.8187307531
tqx = 0.1812692469
e_complete = 50
e_curtate = 49.50166666
var_T = 2500
  by: Constant force of mortality (exponential lifetime)  (t_p_x = e^(-mu t); e° = 1/mu;
e_x = e^(-mu)/(1 - e^(-mu)))
  mu = 0.02
  t = 10
  assumes: 0 < mu < 5; t >= 0
```

What the worked example means: This calculation determines survival probabilities and life expectancies under the assumption that the force of mortality remains constant over time. Given a specific force of mortality 0.02 and a duration of 10 years, the model computes the chance of surviving to that point as 0.818731 and dying before it as 0.181269. It also derives the average total future lifetime, known as the complete expectation, which equals 50, and the average number of full years lived, called the curtate expectation, which is 49.5017. The variance of the future lifetime is calculated as 2500, reflecting the spread of possible death times around the mean.

De Moivre's law of mortality, `act_de_moivre_law`

Survival and death probabilities, the force of mortality, the complete expectation of life and the variance of the future lifetime under de Moivre's law (uniform future lifetime up to the limiting age omega).

$$ {}_tp_x = \frac{\omega - x - t}{\omega - x},\ \mu_{x+t} = \frac{1}{\omega - x - t},\ \mathring{e}_x = \frac{\omega - x}{2} $$

symbol		meaning	SI unit
`omega`	in	limiting age omega (years)	,
`x`	in	age of the life (years)	,
`t`	in	number of years	,
`tpx`	out	probability of surviving t years	,
`tqx`	out	probability of dying within t years	,
`mu`	out	force of mortality at age x+t	,
`e_complete`	out	complete expectation of life e°_x	,
`var_T`	out	variance of the future lifetime	,

Applies when: 0 <= x < omega; 0 <= t < omega - x.

```
$ actuarial-science solve act_de_moivre_law omega=100 x=40 t=10
tpx = 0.8333333333
tqx = 0.1666666667
mu = 0.02
e_complete = 30
var_T = 300
  by: De Moivre's law of mortality  (t_p_x = (omega - x - t)/(omega - x); mu_(x+t) =
1/(omega - x - t); e°_x = (omega - x)/2; Var T = (omega - x)^2/12)
  omega = 100
  x = 40
  t = 10
  assumes: 0 <= x < omega; 0 <= t < omega - x
```

What the worked example means: This calculation applies De Moivre's law of mortality, which assumes a uniform distribution of deaths between the current age and a limiting age. Given an age of 40 and a duration of 10 years up to the limiting age 100, the model computes the probability of surviving t years as 0.833333 and the probability of dying within that period as 0.166667. It also determines the instantaneous rate of death at age x plus t, known as the force of mortality, which equals 0.02. Furthermore, the average remaining lifetime, or complete expectation of life, is calculated as 30, while the variance of the future lifetime is 300.

These results provide a linear framework for estimating mortality risks and life expectancy under the assumption that everyone dies exactly at the limiting age.

Gompertz's law of mortality, `act_gompertz_law`

Survival and death probabilities, the force of mortality at x and x+t and the complete expectation of life under Gompertz's law mu_x = B c^x.

$$\mu_x = Bc^x, \quad {}_tp_x = \exp\left(-\frac{Bc^x(c^t-1)}{\ln c}\right)$$

symbol		meaning	SI unit
`B`	in	Gompertz B (per year)	,
`c`	in	Gompertz c (> 1)	,
`x`	in	age of the life (years)	,
`t`	in	number of years	,
`tpx`	out	probability of surviving t years	,
`tqx`	out	probability of dying within t years	,
`mu_x`	out	force of mortality at age x	,
`mu_xt`	out	force of mortality at age x+t	,
`e_complete`	out	complete expectation of life e°_x	,

Applies when: 0 < B < 0.1; 1 < c < 1.5; 0 <= x <= 120; 0 <= t <= 100.

```
$ actuarial-science solve act_gompertz_law B=0.0003 c=1.07 x=50 t=10
tpx = 0.8813304297
tqx = 0.1186695703
mu_x = 0.008837107519
mu_xt = 0.01738392805
e_complete = 26.69114361
  by: Gompertz's law of mortality  (t_p_x = exp(-B c^x (c^t - 1)/ln c); e°_x = integral of
t_p_x (Gauss-Legendre))
  B = 0.0003
  c = 1.07
  x = 50
  t = 10
  assumes: 0 < B < 0.1; 1 < c < 1.5; 0 <= x <= 120; 0 <= t <= 100
```

In the worked example, B = 0.0003, c = 1.07, x = 50, t = 10 go through the equation above and give tpx = 0.8813304297, tqx = 0.1186695703, mu_x = 0.008837107519, mu_xt = 0.01738392805. It applies only when 0 < B < 0.1; 1 < c < 1.5; 0 <= x <= 120; 0 <= t <= 100: outside that it says what is wrong instead of returning a number.

Makeham's law of mortality, `act_makeham_law`

Survival and death probabilities, the force of mortality at x and x+t and the complete expectation of life under Makeham's law mu_x = A + B c^x.

$$\mu_x = A + Bc^x, \quad {}_tp_x = \exp\left(-At - \frac{Bc^x(c^t-1)}{\ln c}\right), \quad \mathring{e}_x = \int_0^\infty {}_tp_x \, dt$$

symbol		meaning	SI unit
`A`	in	Makeham's age-independent term A (per year)	,
`B`	in	Makeham's B (per year)	,
`c`	in	Makeham's c (> 1)	,
`x`	in	age of the life (years)	,
`t`	in	number of years	,
`tpx`	out	probability of surviving t years	,
`tqx`	out	probability of dying within t years	,
`mu_x`	out	force of mortality at age x	,
`mu_xt`	out	force of mortality at age x+t	,
`e_complete`	out	complete expectation of life e°_x	,

Applies when: 0 <= A < 0.1; 0 < B < 0.1; 1 < c < 1.5; 0 <= x <= 120; 0 <= t <= 100.

```
$ actuarial-science solve act_makeham_law A=0.00022 B=2.7e-06 c=1.124 x=60 t=10
tpx = 0.942549208
tqx = 0.05745079201
mu_x = 0.00322152827
mu_xt = 0.009880632256
e_complete = 27.20968666
  by: Makeham's law of mortality  (t_p_x = exp(-A t - B c^x (c^t - 1)/ln c); e°_x =
integral of t_p_x (Gauss-Legendre))
  A = 0.00022
  B = 2.7e-06
  c = 1.124
  x = 60
  t = 10
  assumes: 0 <= A < 0.1; 0 < B < 0.1; 1 < c < 1.5; 0 <= x <= 120; 0 <= t <= 100
```

What the worked example means: This calculation determines the probability that a person currently aged 60 survives for the next 10 years under Makeham's law, yielding a value of 0.942549. The corresponding probability of death within this period is 0.0574508. Additionally, the force of mortality at the current age is computed as 0.00322153, while the complete expectation of remaining life is estimated to be 27.2097 years. These metrics rely on the specified parameters A, B, and c to model the age-dependent increase in mortality risk.

Questions and reports: support@ai999x.com. AI999X, The Future Is Self-Improving.